When does a tolerant B-rep define a solid? Existence, non-existence and uniqueness of realizations

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Abstract

Boundary representations in CAD kernels carry tolerances: edge curves, the boundary curves of adjacent faces and vertex points do not fit together exactly, and the solid they describe is only implied. We define a realization of a tolerant B-rep as a PL embedding of its cell complex whose cells stay in their tolerance zones, and study existence, non-existence and uniqueness. Uniqueness: if every shell has genus at least one, the carriers of the cells are contractible, and each shell has a product collar containing its zones and a reference surface whose inclusion is a homotopy equivalence, then the images of all realizations are topologically ambient isotopic, and so are the solids they bound. The collar condition follows from a distance bound below the weak feature size of a PL reference surface, for which we give an explicit lower bound in terms of conical radii, with the key lemma machine-checked in Lean. Existence: a conditional fan-and-cone construction for regular complexes, and a re-intersection construction that turns an embedded re-trimmed candidate into a realization. Non-existence: a checkable criterion for crossing planar patches. In OpenCascade 8.0.1, the validity checks decide none of these questions: a sewn wedge passes both checks and has no realization, and a thick pipe that passes both has two realizations whose solids are not homeomorphic.

Keywords: tolerant boundary representation, geometric robustness, realization, ambient isotopy, weak feature size, 3-manifold topology

1. Introduction

1.1. The problem

A solid in a CAD kernel is stored as a boundary representation: a cell complex (faces, edges, vertices) with geometry attached to each cell.

- Each face carries a trimmed patch of a surface.
- Each edge carries a 3D curve, together with one parameter-space curve (pcurve) for every adjacent face.
- Each vertex carries a point.

In floating point, and after operations such as Booleans, fillets, offsets and sewing, these data do not agree. The images of the pcurves on the surfaces miss the 3D edge curve, and the edge curves miss the vertex points. Jackson's local-tolerance approach [12], the published description of Parasolid's, attaches a tolerance to every vertex, edge and face.

The data count as consistent when the deviations lie within the tolerances. OpenCascade, for instance, requires the pcurve images to lie within the edge tolerance of the 3D curve, and the edge ends within the vertex tolerance of the vertex point [15].

What solid does such a model describe? Qi and Shapiro [17] describe the conventional answer: the representation is valid if there exists an exact solid, with the same boundary combinatorial structure, in the tolerance zones. This raises two questions:

1. *Existence.* Is there an embedded surface, with the given cell structure, that respects all the tolerances?
2. *Uniqueness.* If there are several, do they all bound the same solid up to ambient isotopy?

Downstream algorithms (Booleans, offsets, meshing, analysis) implicitly assume both answers are yes.

1.2. What we prove

We call an embedding as in (1) a *realization*: a PL embedding h of the cell complex K with $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for every cell σ, where $\mathcal{Z}(\sigma)$ is the tolerance zone of σ (Section 2).

- **Uniqueness** (Theorem 3.5).
  - *Hypotheses:* the carriers of the cells are contractible; each shell has a product collar that contains its zones and a carried reference surface whose inclusion is a homotopy equivalence; the collars are pairwise disjoint; and every shell has genus at least one.
  - *Conclusion:* the *images* of any two realizations are ambient isotopic, and so are the solids they bound.
  - *Method:* a carrier lemma (a cellular homotopy argument), combined with Waldhausen's theorem that incompressible surfaces in a product $F \times I$ can be made horizontal [23, Prop. 3.1].
- **The collar condition** (Section 4).
  - The collar condition follows from a distance bound below the weak feature size $\operatorname{wfs}(M)$ of a reference surface M (Lemma 4.1). One step uses Moise's approximation of homeomorphisms by PL homeomorphisms [14, Thm. 35.2].
  - For a polyhedral M we prove $\operatorname{wfs}(M) \geq \lambda(M) > 0$, where λ is defined from the conical radii of M (Corollary 4.3). The underlying cone lemma is machine-checked in Lean 4 with Mathlib.
  - We compute lower bounds for $\operatorname{wfs}(M)$ with an octree search, up to stated floating-point qualifications.
- **Existence** (Section 5). There are two constructions:
  - fans in the normal disks of the edges, with cones at the vertices. This is a conditional theorem for regular complexes (Theorem 5.1), proved in Appendix Appendix B. Its hypotheses are local conditions on edges and vertices, global disjointness conditions (S1), and two non-local topological conditions (V1(b), P0).

  - re-intersection of the face surfaces, for sharp edges with tight tolerances (Propositions 5.3 and 5.4; the curved case is proved in Appendix Appendix C). These assume that the re-trimmed candidate is embedded and lies in the zones, and turn it into a PL realization.

- **Non-existence** (Theorem 6.1). Under explicit conditions, if two planar patches cross transversally away from the projections ("shadows") of all edge and vertex zones, there is no realization.

1.3. What the experiments show

In OpenCascade 8.0.1 (Section 7):

- `BRepCheck_Analyzer` accepts models however large their tolerances are; in the source, its tolerance-value check is declared but not implemented.

- `BRepAlgoAPI_Check` gives false alarms on realizable models, and accepts models whose tolerances exceed the weak feature size of the nominal surface. Among them is a thick pipe with two realizations whose solids are not homeomorphic: one ties a trefoil into the hole.

- Sewing a wedge whose faces are slightly displaced produces a model that passes both checks and, by Theorem 6.1, has no realization. The hypotheses were verified in exact rational arithmetic.

1.4. Related work

Tolerant modelling

- Segal [20] assigned a tolerance to every vertex, edge and face of polyhedral models.

- Jackson [12] gave the local-tolerance B-rep approach.

- Milenkovic [13] proved, in the plane and with a uniform ε, that normalized data have an exact realization within $\varepsilon/2$.

- Sakkalis, Shen and Patrikalakis [18] study interval solids: boxes covering the boundary of a known exact solid, up to homeomorphism. Sakkalis, Peters and Bisceglio [19, Thm. 10] prove ambient isotopy for a closed surface that lies inside a normal tube of a smooth closed surface and meets each normal fibre once. The tube radius there is bounded by curvature and by a double-normal distance.

- Qi and Shapiro [16] develop ε-solidity, which does not require an exact solid to exist [17].

- Shapiro [21] gives conditions under which the union of the tolerance zones is homotopy equivalent to the cell complex. In §5.2 (p. 178) he notes that the homotopy condition alone is not sufficient, since a valid boundary representation must also be an orientable 2-cycle separating space. It is "not immediately clear" how to extend these notions to unions of tolerance zones, though the notion of separation used in isotopy results is likely to be useful. He also treats the containment of the exact cells in their zones as a much stronger demand. Realizations satisfy exactly that containment.

Constructing an exact solid

- Andersson, Stewart and Zidani [2] and Stewart and Zidani [22] construct a solid from inconsistent trimmed-patch data by perturbation (Whitney extension, transfinite interpolation). They *assume* that the perturbed patches embed:
  - ASZ say that adjacent patches need an a posteriori check [2, pp. 819–820];
  - SZ make a similar assumption [22, p. 188], and derive a normal-vector bound intended to exclude extraneous intersections.
- In an earlier paper [3], the same authors give checkable conditions under which a single composite parametric surface on a common parameter domain is injective, by a degree argument. They note that separately trimmed B-rep faces would need extra inside/outside information [3, §3.7].

Isotopy

- The weak feature size and the isotopy of offsets below it are studied by Chazal and Lieutier [7] and by Chazal, Cohen-Steiner and Lieutier [5]. The latter also prove isotopy of offset level sets of general compact sets under Hausdorff perturbation [6, Thms. 4.2, 4.3].
- Chazal and Cohen-Steiner [4, Thm. 2.1] prove that a surface homeomorphic to S, lying in a thickening of S and separating its sides, is isotopic to S. This is close to the topological core of our uniqueness argument.
- Andersson, Peters and Stewart [1, Thm. 1] prove that a one-parameter family of properly joined curvilinear triangular patches, each injective with non-singular Jacobian and adjacent patches nontangential at every time, is linked by an ambient isotopy. Their nontangentiality is close to our no-fold clause T in Proposition 5.4. The settings differ: they assume the perturbed complex is properly joined at every time, while we start from data whose patches do not meet and construct a realization.

1.5. What is new and what is not

For each result we say what was known, what we add, and how far we looked. "We did not find" refers to the literature cited in this paper and the searches behind it, not to an exhaustive survey.

Uniqueness (Theorem 3.5)

- *Known.* The topological core is classical. Waldhausen [23, Prop. 3.1] shows that incompressible surfaces in $F \times I$ can be made horizontal. Chazal and Cohen-Steiner [4, Thm. 2.1] prove that a surface homeomorphic to S, lying in a thickening of S and separating its sides, is isotopic to S. Sakkalis, Peters and Bisceglio [19, Thm. 10] prove isotopy for a surface in a normal tube of a smooth surface that meets each fibre once.
- *What we add.*
  - The statement for tolerant B-reps: per-cell zones, and realizations that keep every cell in its own zone.

  - The carrier lemma (Lemma 3.1). With contractible carriers it makes every realization homotopic, inside the collar, to the reference surface. Together with the collar hypothesis that the reference surface's inclusion is a homotopy equivalence, this gives the π_1-isomorphism needed for Waldhausen's theorem (Lemma 3.2). So a hypothesis is still needed in place of separation, namely the homotopy-equivalence condition, but only for the single reference surface, not for every realization as in [4], and no realization needs to meet each fibre once as in [19].
  - A collar from the weak feature size (Lemma 4.1), for PL reference surfaces. These may have sharp edges and triangulated small fillets. The result of [19] is stated for a smooth closed reference surface, with the tube radius bounded by curvature and by a double-normal distance. Each result has its own setting restriction: ours needs a PL reference.

- *Limits.*
  - The collar condition is a hypothesis. Lemma 4.1 supplies it only when the reference surface is a carried PL embedding of the shell, every zone lies in $N_d(M)$ with $d < r < \mathrm{wfs}(M)$, the closed r-neighbourhoods of different shells are pairwise disjoint, and the genus is at least one.
  - Shells of genus 0 are excluded.
  - The isotopy is topological, and of images, not of embeddings.

The explicit bound $\mathrm{wfs}(M) \geq \lambda(M)$ *(Corollary 4.3)*

- *Known.* That $\mathrm{wfs}(M) > 0$ for compact semialgebraic M. Fu [11, §5.3] proves that the distance function of a compact semialgebraic set has finitely many critical values. di Rocco, Eklund and Gäfvert [8, Thm. 2.5] state positivity and attribute it to Fu. Chazal and Lieutier [7, Prop. 3.6] state finiteness for bounded subanalytic open sets, attribute it to Fu for semialgebraic sets, and refer to an earlier paper of theirs for the adaptation. Fu defines semialgebraic sets by one finite system of inequalities; his argument (Tarski–Seidenberg, a finite decomposition into smooth submanifolds, his Lemma 4.3 and Sard's theorem) does not use that restriction; that it covers polyhedral surfaces is our reading.

- *What we add.* An explicit lower bound in terms of conical radii, whose key cone lemma is machine-checked in Lean 4; the positivity $\lambda(M) > 0$ is proved on paper and not formalized. We did not find such a bound. That does not make it new.

Existence (Theorem 5.1, Propositions 5.3 and 5.4)

- *Known.*
  - Milenkovic [13]: existence in the plane, with one uniform tolerance.
  - Andersson, Stewart and Zidani [3]: checkable injectivity conditions for one composite patch on a common parameter domain.
  - [2, 22]: they construct a solid from inconsistent data but assume that the perturbed patches embed.

  - Andersson, Peters and Stewart [1, Thm. 1]: ambient isotopy along a family that stays properly joined. Their nontangentiality is close to our no-fold clause T.

- *What we add.* Sufficient conditions under which a 3D B-rep with per-cell tolerance zones has a realization of the whole cell complex. They are of two different kinds.
  - Theorem 5.1 constructs the realization from conditions on the data. It is proved for regular complexes only, and it is conditional: two of its hypotheses are non-local.
  - Propositions 5.3 and 5.4, for sharp edges with tight tolerances, assume that a re-trimmed candidate surface is already embedded and lies in the zones (in Proposition 5.4, clauses C and B1–B3), and show that it gives a realization; for curved faces this needs a PL approximation. They check a given candidate rather than derive embeddedness. Proposition 5.3 is for planar faces and vertices of degree three; Proposition 5.4 needs C^2 regular face parametrizations, without poles or cone apices. On curved models these clauses are checked on samples in double precision, and the vertex part of clause T is not checked.

- *How far we looked.* We did not find existence results of either kind for 3D B-reps with per-cell tolerance zones.

- *Not covered.* Sewn models with tangent edges and OpenCascade's vertex tolerances. Whether they have realizations is open (Section 8).

Non-existence (Theorem 6.1)

- *What we add.* A checkable criterion for planar patches. We apply it to sewn models that pass both of OpenCascade's checks.

- *Limits.* The theorem concerns one pair of planar faces whose closed cells are disks bounded by single wires, with simply connected edge zones (O3). A version for curved faces is open.

- *How far we looked.* We did not find a non-existence criterion for tolerant B-reps in the literature we read.

Experiments (Section 7)

- *What we add.* The observation that OpenCascade's validity checks decide neither existence nor uniqueness.

- *Scope.* The evidence is a handful of parts in one kernel version (8.0.1). We did not test ACIS or Parasolid.

2. Tolerant B-reps and realizations

2.1. Complexes

K is a finite cell complex in the B-rep sense. Its 0-cells are points, its edges are arcs or closed curves, and its faces are compact surfaces, possibly with holes. The open face is

the face minus all its edges, seams included. The underlying space $|K|$ is a disjoint union of closed connected surfaces $|K_1|, \dots, |K_n|$ (the *shells*). We write $\tau \le \sigma$ when τ is a face of σ (including $\tau = \sigma$). For the uniqueness theorem we assume K is *regular*: every closed cell $\bar{\sigma}$ is a ball whose boundary is a subcomplex. B-rep complexes with seam edges, closed edges or faces with holes are not regular, and must first be subdivided. A new cell inherits the zone of the open cell of K that contains it. This keeps the zones monotone, and a realization of the original model is a realization of the subdivision, since $h(\operatorname{int}\sigma') \subseteq h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for a new cell $\sigma' \subseteq \sigma$. The carrier condition of Section 3 must then hold for the subdivided complex. A new edge inside a face f has carrier $\mathcal{Z}(f)$, so subdividing faces adds the hypothesis $\pi_1(\mathcal{Z}(f)) = 0$; contractible zones satisfy it. A map of $|K|$ is *PL* if it is linear on the simplices of some triangulation of $|K|$ that subdivides its cells. In dimension two, any two such triangulations are related by a cell-preserving PL homeomorphism (standard; not separately checked against a source), so the notion does not depend on the choice. New vertices created inside an edge or face inherit its zone, so the carrier condition requires those zones to be path-connected. The existence results say where they apply to B-rep complexes directly.

2.2. Geometric data and zones

- Each vertex v has a point p_v and a tolerance τ_v.
- Each edge e has a C^2 curve $\gamma_e\colon [0,1] \to \mathbb{R}^3$ and a tolerance τ_e.
- Each face f has a C^2 surface S_f, a compact trimmed patch $P_f \subseteq S_f$, and a tolerance τ_f. The boundary of P_f consists of the images $\beta_{f,e} = S_f \circ (\text{pcurve})$ of the pcurves of the edges $e < f$.

The *deviation* of e is $\operatorname{dev}_e = \max_{f>e} \max_t |\beta_{f,e}(t) - \gamma_e(t)|$, measured at equal parameters. OpenCascade's SameParameter property asks for $\operatorname{dev}_e \le \tau_e$ in this sense; its format specification defines the edge tolerance as a one-sided distance from the pcurve images to the 3D curve [15].

Definition 2.1 (Zones)**.** We use *augmented* open zones:

- $\mathcal{Z}(v) = B(p_v, \tau_v)$;
- $\mathcal{Z}(e) = N_{\tau_e}(\gamma_e) \cup \bigcup_{v<e} \mathcal{Z}(v)$;
- $\mathcal{Z}(f) = N_{\tau_f}(P_f) \cup \bigcup_{e<f} \mathcal{Z}(e)$.

Here N_r is the open r-neighbourhood. Augmentation makes $\mathcal{Z}$ monotone: $\tau \le \sigma$ implies $\mathcal{Z}(\tau) \subseteq \mathcal{Z}(\sigma)$.

Definition 2.2 (Realization)**.** A *realization* of the tolerant B-rep $T = (K, \text{data}, \tau)$ is a PL embedding $h\colon |K| \to \mathbb{R}^3$ with $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for every cell σ.

Remark 2.3.
- *Open-cell condition.* We require only $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$. The closed-cell condition with unaugmented neighbourhoods, $h(\bar{\sigma}) \subseteq N_{\tau_\sigma}(\cdot)$, usually fails for every candidate, mainly because face tolerances are tiny: $h(v)$ would have to lie within τ_f of every incident patch. With augmented zones the open-cell condition implies $h(\bar{\sigma}) \subseteq \mathcal{Z}(\sigma)$, since $\bar{\sigma}$ is the union of the open cells $\tau \le \sigma$.

- *Tameness.* Requiring h to be PL rules out wild embeddings such as Alexander's horned sphere.

Definition 2.4 (Carriers)**.** The *carrier* of a cell is $\mathcal{C}(\sigma) = \bigcup_{\tau \le \sigma} \mathcal{Z}(\tau)$. For augmented zones $\mathcal{C}(\sigma) = \mathcal{Z}(\sigma)$. A continuous map $g\colon |K| \to \mathbb{R}^3$ is *carried* if $g(\bar{\sigma}) \subseteq \mathcal{C}(\sigma)$ for every σ. By Remark 2.3, every realization is carried.

Definition 2.5 (Solid)**.** Let $S_1, \dots, S_n \subset \mathbb{R}^3$ be pairwise disjoint closed surfaces. The *depth* of a point $x \notin \bigcup S_j$ is the number of j for which x lies in the bounded component of $\mathbb{R}^3 \setminus S_j$. The solid bounded by the S_j is the closure of the set of points of odd depth.

This matches the usual B-rep convention of outer shells and cavities when the shells are consistently oriented; a consistency check of the orientation is separate.

3. Uniqueness for shells of positive genus

3.1. Hypotheses

- **Carrier condition.** $\mathcal{C}(v)$ is path-connected for every vertex v, and $\mathcal{C}(\sigma)$ is path-connected with $\pi_k(\mathcal{C}(\sigma)) = 0$ at every base point, for every k-cell σ with $k \ge 1$. Contractible carriers suffice.

- **Collar condition.** For each shell K_i there are a compact PL submanifold $U_i \subset \mathbb{R}^3$, a closed surface M_i, a PL homeomorphism $\Psi_i\colon U_i \to M_i \times [-1, 1]$, and a carried PL embedding $\varphi_i\colon |K_i| \to \mathbb{R}^3$ (a *reference*), with these properties:

  - all zones of the cells of K_i lie in $\operatorname{int} U_i$;
  - $\varphi_i(|K_i|) \subseteq \operatorname{int} U_i$, and the inclusion $\varphi_i(|K_i|) \hookrightarrow U_i$ is a homotopy equivalence;
  - the U_i are pairwise disjoint.

  (The reference need not be a level $\Psi_i^{-1}(M_i \times \{t\})$. This weaker form is what Lemma 4.1 can deliver.) Note that M_i is connected, since $M_i \simeq U_i \simeq |K_i|$; it is orientable, since $M_i \times I \cong U_i \subset \mathbb{R}^3$; and it is not a sphere when $|K_i|$ is not, since then $\pi_1(M_i) \cong \pi_1(|K_i|) \neq 1$.

- **Genus.** Each $|K_i|$ has genus at least one.

3.2. Lemmas

Lemma 3.1 (Carrier lemma)**.** *Under the carrier condition, any two carried maps $f, g\colon |K| \to \mathbb{R}^3$ are homotopic by a homotopy H with $H(\bar{\sigma} \times I) \subseteq \mathcal{C}(\sigma)$ for every σ.*

Proof. Define H on $|K| \times \{0, 1\} \cup |K^{(k)}| \times I$ by induction on k, keeping $H(\bar{\sigma} \times I) \subseteq \mathcal{C}(\sigma)$, with $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$.

- $k = 0$: for a vertex v, the points $f(v), g(v)$ lie in the path-connected set $\mathcal{C}(v)$; join them by a path in $\mathcal{C}(v)$.

- $k \ge 1$: let σ be a k-cell. $\bar{\sigma} \times I$ is a $(k+1)$-ball whose boundary $\bar{\sigma} \times \{0, 1\} \cup \partial\bar{\sigma} \times I$ is a k-sphere, and H is already defined on it:

– on $\bar{\sigma} \times \{0,1\}$ it is f or g, with image in $\mathcal{C}(\sigma)$;

– on $\partial\bar{\sigma} \times I$ it is defined by induction, with image in $\bigcup_{\tau<\sigma} \mathcal{C}(\tau) \subseteq \mathcal{C}(\sigma)$.

Since $\pi_k(\mathcal{C}(\sigma)) = 0$, H extends over $\bar{\sigma} \times I$ with image in $\mathcal{C}(\sigma)$.

The pieces agree on common boundaries, and a finite union of closed sets glued continuously gives a continuous H. □

Lemma 3.2 (Separation)**.** *Assume the carrier and collar conditions, and let h be a realization. Then $G = h(|K_i|)$ is a closed surface in* $\operatorname{int} U_i$*, and the inclusion $G \hookrightarrow U_i$ is a homotopy equivalence.*

Proof.

- By Lemma 3.1 on K_i, h and φ_i are homotopic through maps with image in $\bigcup \mathcal{C}(\sigma) \subseteq \operatorname{int} U_i$, so the homotopy takes place in U_i.
- $\varphi_i \colon |K_i| \to U_i$ is a homeomorphism onto $\varphi_i(|K_i|)$ followed by the inclusion, which is a homotopy equivalence by the collar condition. Hence so is $h \colon |K_i| \to U_i$.
- Since h is a homeomorphism onto G, the inclusion $G \hookrightarrow U_i$, which equals $h \circ h^{-1}$, is a homotopy equivalence.

□

Lemma 3.3 (Horizontality)**.** *Let $U \cong_\Psi M \times [-1,1]$ be as in the collar condition, with M not a sphere, and let $G \subset \operatorname{int} U$ be a closed PL surface whose inclusion into U is a homotopy equivalence. Then there is an ambient isotopy of $\mathbb{R}^3$, supported in U and fixed on ∂U, that carries G to $\Psi^{-1}(M \times \{0\})$.*

Proof. (a) *G is incompressible.*

- By hypothesis, $\pi_1(G) \to \pi_1(U)$ is an isomorphism.
- The boundary of a compressing disk would be a curve that is essential in G and null-homotopic in U, which this rules out. That is incompressibility in the sense of Waldhausen [23, p. 58].
- $G \simeq U \simeq M$, so G is not a sphere and case (b) of that definition does not arise. We do not use the converse direction, which needs the loop theorem.
- G is a closed orientable surface in an orientable 3-manifold, hence two-sided, and it is disjoint from ∂U; so it satisfies Waldhausen's conventions [23, §0].

(b) *G can be made horizontal.*

- Apply Waldhausen's Proposition 3.1 [23, pp. 65–68] to $U \cong M \times I$ (M orientable, not a sphere) and the incompressible surface G. Since $\partial G = \emptyset$, the condition "$\partial G \subseteq F \times 1$" holds vacuously.
- It gives an isotopy, constant on ∂U, from G to a surface G' whose projection $p \colon G' \to M$ is a homeomorphism onto its image.
- $p(G')$ is open (invariance of domain) and closed in the connected surface M, so $p(G') = M$. Hence G' is the graph of a function $s \colon M \to (-1,1)$.

- Waldhausen's proof of Proposition 3.1 uses that $F \times I$ is irreducible, explicitly in the base case (F a disc, annulus or pair of pants, p. 66) and, through the induction on p. 67, for compact surfaces F' with boundary obtained by cutting. Lemma 3.4 below covers all these cases.

(c) *Flatten.* The PL homeomorphisms of the fibres $\{x\} \times [-1, 1]$ that fix both ends and send $s(x)$ to $(1-t)s(x)$, $t \in [0, 1]$, give an ambient isotopy of U, fixed on ∂U, from G' to $\Psi^{-1}(M \times \{0\})$. This isotopy is topological.

(d) The isotopy of (b) is already ambient: in Waldhausen's paper an isotopy of subspaces Z_1, Z_2 of X is by definition an isotopy of the identity map of X carrying Z_1 to Z_2 [23, p. 57], and Proposition 3.1 gives one constant on ∂U. Both isotopies are fixed on ∂U and extend by the identity to $\mathbb{R}^3$.

□

Lemma 3.4. *Let F be a compact orientable PL surface other than the 2-sphere. Then $F \times I$ is irreducible: every PL 2-sphere $S \subset \mathrm{int}(F \times I)$ bounds a PL 3-cell in $F \times I$. (Following Waldhausen's convention that surfaces are properly embedded, a 2-sphere in $F \times I$ is disjoint from the boundary.)*

Proof. Case $\partial F \neq \emptyset$.

- Then $F \times I$ is a handlebody: a regular neighbourhood H of a finite graph in $\mathbb{R}^3$, with ∂H connected (standard; not separately checked against a source). We work with $H \subset \mathbb{R}^3$.

- By the PL Schoenflies theorem [14, Thm. 17.12], S bounds a PL 3-cell $B \subset \mathbb{R}^3$. The set $\mathbb{R}^3 \setminus H$ is connected (Alexander duality, as ∂H is connected), unbounded and disjoint from S, so it lies in the unbounded component of $\mathbb{R}^3 \setminus S$. Hence $B \subseteq H$.

Case F closed, of genus at least one.

- Let $q\colon \tilde{F} \times I \to F \times I$ be the universal covering, with deck group Γ acting by $\gamma \times \mathrm{id}$. $\tilde{F}$ is a simply connected non-compact PL surface without boundary, so it is PL homeomorphic to $\mathbb{R}^2$ (standard; not separately checked against a source). Hence $\tilde{F} \times I \cong_{PL} \mathbb{R}^2 \times I \subset \mathbb{R}^3$.

- S is simply connected, so it lifts to a PL 2-sphere $\tilde{S} \subset \mathbb{R}^2 \times (0, 1)$, and the lifts $\gamma\tilde{S}$, $\gamma \in \Gamma$, are pairwise disjoint. By the PL Schoenflies theorem [14, Thm. 17.12], $\tilde{S}$ bounds a PL 3-cell $\tilde{B} \subset \mathbb{R}^3$. Points of $\mathbb{R}^3$ outside $\mathbb{R}^2 \times [0, 1]$ are joined to infinity in the complement of $\tilde{S}$, so $\tilde{B} \subset \mathbb{R}^2 \times I$.

- Let $\gamma \neq 1$. The spheres $\tilde{S}$ and $\gamma\tilde{S}$ are disjoint, so the cells $\tilde{B}$ and $\gamma\tilde{B}$ are either disjoint or nested. If $\gamma\tilde{B} \subseteq \tilde{B}$, then $\gamma^n\tilde{B} \subseteq \tilde{B}$ for all $n \geq 1$. Γ is torsion-free, so the γ^n are distinct, and infinitely many deck transformations move the compact set $\tilde{B}$ to meet itself. That contradicts proper discontinuity. The case $\tilde{B} \subseteq \gamma\tilde{B}$ is the same with γ^{-1}.

- So the translates of $\tilde{B}$ are pairwise disjoint, q is injective on $\tilde{B}$, and $q(\tilde{B})$ is a PL 3-cell in $F \times I$ with boundary S.

Also standard and not separately checked: that π_1 of a closed surface is torsion-free, and that deck groups act properly discontinuously. □

3.3. The theorem

Theorem 3.5 (Uniqueness)**.** *Assume the carrier condition, the collar condition, and that every shell has genus at least one. Then:*

1. *the image of any realization is ambient isotopic (by a topological isotopy) to* $\bigcup_i \Psi_i^{-1}(M_i \times \{0\})$*;*
2. *the images of any two realizations are ambient isotopic;*
3. *the solids they bound (Definition 2.5) are ambient isotopic.*

Proof.

- By Lemma 3.2, each $G_i = h(|K_i|)$ satisfies the hypotheses of Lemma 3.3 in U_i. Apply that lemma shell by shell. The supports U_i are pairwise disjoint, so the isotopies Φ^i can be run simultaneously; their end map is a homeomorphism Φ of $\mathbb{R}^3$.
- $\Phi(G_j) = \Psi_j^{-1}(M_j \times \{0\})$ for every j: on U_j, every Φ^i with $i \neq j$ is the identity, because its support U_i is disjoint from U_j.
- By the Jordan–Brouwer separation theorem each closed surface separates $\mathbb{R}^3$ into a bounded and an unbounded component. A homeomorphism of $\mathbb{R}^3$ maps sets with compact closure to sets with compact closure, so Φ maps the bounded component of $\mathbb{R}^3 \setminus G_j$ onto the bounded component of $\mathbb{R}^3 \setminus \Phi(G_j)$. So depth is preserved, and Φ maps the solid bounded by the G_j onto the solid bounded by the levels.
- For two realizations with maps Φ and Φ', use $\Phi'^{-1} \circ \Phi$.

□

Remark 3.6.

- *What is not covered.* Spheres (genus 0) are excluded, and the isotopy is of images, not of embeddings.
- *Relation to Chazal–Cohen-Steiner.* Chazal and Cohen-Steiner [4, Thm. 2.1] prove that a compact orientable surface $S' \cong S$ lying in a thickening of S and separating its sides is isotopic to S in the thickening. Lemmas 3.2 and 3.3 play the same role here. What is specific to tolerant B-reps is the set-up: per-cell zones, cell-respecting realizations, the carrier lemma, and π_1 in place of separation, which the carrier lemma provides.
- *The collar condition cannot be dropped.* The thick pipe of Section 7 passes both OpenCascade checks and has two realizations whose solids are not even homeomorphic.
- *Standard facts used without separate checking against a source:* continuity of cell-by-cell extension, orientability of closed surfaces in $\mathbb{R}^3$, invariance of domain, the Jordan–Brouwer separation theorem, and that the fibrewise flattening is an ambient isotopy.

4. From the weak feature size to the collar condition

4.1. Weak feature size

Let $M \subset \mathbb{R}^3$ be compact, $d(p) = \operatorname{dist}(p, M)$, and let $\Gamma(p)$ be the set of nearest points of p in M. A point $p \notin M$ is *critical* if $0 \in \operatorname{conv}\{p - x : x \in \Gamma(p)\}$. The *weak feature size* $\operatorname{wfs}(M)$ is the infimum of the positive critical values of d [5, §2.2]; see also [7]. Unlike the reach, it stays positive at sharp edges and small fillets. A surface with a concave edge has reach 0, and a fillet of radius r caps the reach at r.

Lemma 4.1 (Collar)**.** *Let M be a closed connected PL surface in $\mathbb{R}^3$, and let $0 < d < r <$ $\operatorname{wfs}(M)$.*

(a) The closed neighbourhood $\bar{N}_r(M)$ is homeomorphic to $M \times [-1,1]$, with M going to $M \times \{0\}$.

(a′) There is a compact PL submanifold U with $\bar{N}_d(M) \subseteq \operatorname{int} U$, $U \subseteq \operatorname{int} \bar{N}_r(M)$ and $U \cong_{PL} M \times [-1,1]$.

(b) Suppose M has genus at least one, and let U be as in (a′). Suppose further that $M = \varphi(|K_i|)$ for a carried PL embedding φ of a shell, that every zone of a cell of that shell lies in $N_d(M)$, and that the sets $\bar{N}_r(M_i)$ of different shells are pairwise disjoint. Then the collar condition of Section 3 holds, with $U_i = U$.

Proof. Write $d = d_M$. By the Jordan–Brouwer separation theorem, $\mathbb{R}^3 \setminus M$ has two components, a bounded one Ω_- and an unbounded one Ω_+.

(a) *Step 1: small radii.*

- Work in a large simplicial cube Q containing $\bar{N}_r(M)$, triangulated so that M is a subcomplex. The restriction of d to Q is semialgebraic (Tarski–Seidenberg), proper, nonnegative, with zero set the compact set M. Durfee's definition of a (semialgebraic) *rug function* [9, Def. 3.1] asks exactly this and no smoothness, so $d|_Q$ is a rug function for M in Q. For suitably small $\delta > 0$, the set $\bar{N}_\delta(M) = d^{-1}[0,\delta]$ is therefore a semialgebraic neighbourhood of M in Q [9, Def. 3.4].

- A PL regular neighbourhood R of M in Q is also a semialgebraic neighbourhood [9, Prop. 3.7]. By the uniqueness of semialgebraic neighbourhoods [9, Prop. 3.5] there are homeomorphisms $h_t \colon Q \to Q$ with $h_t|_M = \mathrm{id}$ for all t and $h_1(\bar{N}_\delta(M)) = R$. So h_1 is a homeomorphism of pairs $(\bar{N}_\delta(M), M) \to (R, M)$ that fixes M pointwise. We take $\delta < r$.

- M is a closed orientable surface in $\mathbb{R}^3$, hence two-sided, so $R \cong M \times [-1,1]$ with M going to $M \times \{0\}$. Hence $(\bar{N}_\delta(M), M) \cong (M \times [-1,1], M \times \{0\})$. Its frontier is $S_\delta = \{d = \delta\}$, because δ is not a critical value, so every point with $d = \delta$ is a limit of points with $d > \delta$ (the directional-derivative argument of Step 3). S_δ has two components, the images of $M \times \{\pm 1\}$, and they lie in different components of $\mathbb{R}^3 \setminus M$. The two halves $M \times (0,1]$ and $M \times [-1,0)$ are connected, so each lies in one of $\Omega_\pm$. Every neighbourhood of a point of $M = \partial\Omega_+ = \partial\Omega_-$ meets both, so the halves lie in different ones. Write $S_\delta^\pm = S_\delta \cap \Omega_\pm$.

Step 2: the two shells. Let $A_\pm = \{x \in \Omega_\pm : \delta \le d(x) \le r\}$.

- *Inner side.* Put $O = \Omega_-$. It is bounded and open, and for $x \in O$ the distance to O^c equals $d(x)$, with the same nearest points (they lie on $\partial O = M$). So the critical points of the distance to O^c in O are exactly the critical points of d in Ω_-. The criticality notions agree: [7] use the vanishing of their gradient field, which holds iff x lies in the convex hull of its nearest boundary points [7, Lemma 2.2] (see also [5, §2.2]).

- Since $r < \operatorname{wfs}(M)$, the shell A_- contains no critical point. The isotopy lemma [7, Prop. 3.4], which they reproduce from Grove's critical point theory for distance functions, gives a homeomorphism $A_- \cong S_\delta^- \times [\delta, r]$.

- *Outer side.* Put $O = \Omega_+ \cap B(0, \mathcal{R})$ with $\mathcal{R}$ large. For $x \in O$ the distance to O^c is $\min(d(x), \mathcal{R} - |x|)$. Take $\mathcal{R}$ larger than $2r$ plus the largest $|y|$, $y \in M$. Then the points of O with distance to O^c in $[\delta, r]$ form two disjoint pieces: A_+, where $d < \mathcal{R} - |x|$ strictly, so the distance equals d with the same nearest points, and a shell near the sphere $|x| = \mathcal{R}$, where it equals $\mathcal{R} - |x|$.

- Neither piece contains a critical point. In A_+ this is because $r <$ wfs. Near the sphere the only nearest point is the radial projection.

- So Prop. 3.4 gives a homeomorphism of the union with $R^{-1}(\delta) \times [\delta, r]$, where $R^{-1}(\delta) = S^+_\delta \sqcup \{|x| = \mathcal{R} - \delta\}$ has exactly two components ($S^+_\delta \cong M$ is connected). The components of the product are $C \times [\delta, r]$ for the two components C. The shell near the sphere is one component of the union, so A_+ is the other: $A_+ \cong C \times [\delta, r]$ with C connected. Step 3 shows that $C \cong S^+_\delta$.

- We use Prop. 3.4 in its stated closed-shell form: for a bounded open O and $R = d(\cdot, O^c)$, if the shell $\{x \in O : \rho_1 \le R(x) \le \rho_2\}$ contains no critical point of R, it is homeomorphic to $R^{-1}(\rho_1) \times [\rho_1, \rho_2]$ [7, Prop. 3.4].

Step 3: gluing.

- *The boundary of $A_\pm$.* $A_\pm$ is a compact 3-manifold in $\mathbb{R}^3$, so by invariance of domain its manifold boundary is its frontier in $\mathbb{R}^3$.

  - Points with $\delta < d < r$ are interior points.
  - A point with $d = \delta$ is a limit of points with $d < \delta$: move along the segment to a nearest point.
  - A point x with $d = r$ is a limit of points with $d > r$. Since x is not critical, there is a unit vector u with $\langle u, x - y\rangle > 0$ for all $y \in \Gamma(x)$. The one-sided derivative of d in direction u is $\min_{y \in \Gamma(x)} \langle u, (x - y)/|x - y|\rangle > 0$ (standard formula for the directional derivative of a distance function; not separately checked against a source).

  So $\partial A_\pm = S^\pm_\delta \sqcup S^\pm_r$, and the connected closed set $S^\pm_\delta$ is one boundary component. Every boundary component of $C \times [\delta, r]$ is a copy of C, so $C \cong S^\pm_\delta$.

- A homeomorphism between manifolds with boundary maps boundary components to boundary components. So the homeomorphism of Step 2 takes $S^\pm_\delta$, a boundary component of $A_\pm$, onto one end of the product. Composing with a flip of $[\delta, r]$ if necessary, and with the inverse of its restriction to $S^\pm_\delta$, we get $A_\pm \cong S^\pm_\delta \times [0, 1]$ with $S^\pm_\delta$ going to $S^\pm_\delta \times \{0\}$.

- Then $\bar{N}_r(M) = \bar{N}_\delta(M) \cup A_+ \cup A_-$ is $M \times [-1, 1]$ with a product collar attached to each boundary component along that component. That is homeomorphic to $M \times [-2, 2] \cong M \times [-1, 1]$, with M still going to $M \times \{0\}$.

(a′) The interior of $\bar{N}_r(M)$ is $N_r(M)$: by the computation in Step 3, a point with $d = r$ is a limit of points with $d > r$.

- By (a), there is a homeomorphism $g\colon M \times (-1,1) \to N_r(M)$.
- $g^{-1}(\bar{N}_d(M))$ is compact, so it lies in $M \times [-a,a]$ for some $a < 1$. Take $a < b < 1$ and $K = M \times [-b,b]$, a polyhedral 3-manifold with boundary in the PL 3-manifold $M \times (-1,1)$.
- The compact set $g(K)$ and the closed set $\mathbb{R}^3 \setminus N_r(M)$ are disjoint, and so are the compact sets $g(\partial K)$ and $\bar{N}_d(M)$. Let $\varepsilon > 0$ be smaller than both distances. By Moise's approximation theorem [14, Thm. 35.2] there is a PL homeomorphism f of K into $\mathbb{R}^3$ with $|f - g| < \varepsilon$ on K. Put $U = f(K)$.
- Then U is a compact PL submanifold, $U \cong_{PL} M \times [-1,1]$, and $U \subseteq N_r(M) = \operatorname{int} \bar{N}_r(M)$.
- $\bar{N}_d(M) \subseteq \operatorname{int} U$. Let $x \in \bar{N}_d(M)$. The straight-line homotopy from $g|_{\partial K}$ to $f|_{\partial K}$ avoids x, so each of the two boundary surfaces has x in its bounded complementary component for f exactly when it does for g (the mod-2 degree of the radial projection from x is a homotopy invariant). For a compact connected 3-manifold in $\mathbb{R}^3$ with two boundary surfaces, the interior is the set of points of depth one with respect to them (Definition 2.5; standard, by Alexander duality). Since $x \in g(\operatorname{int} K)$, its depth is one for g, hence for f, so $x \in \operatorname{int} U$.
- f need not send M to a level, which is why the collar condition of Section 3 does not ask for that.

(b) The zones lie in $N_d(M) \subseteq \operatorname{int} U$, and $M \subseteq N_d(M) \subseteq \operatorname{int} U$. The U_i are pairwise disjoint because they lie in the disjoint sets $\bar{N}_r(M_i)$. It remains to show that the inclusion $M \hookrightarrow U$ is a homotopy equivalence.

- By (a), the composite $M \hookrightarrow U \hookrightarrow \bar{N}_r(M)$ is a homotopy equivalence. So $\pi_1(U) \to \pi_1(\bar{N}_r(M))$ is onto.
- Both groups are isomorphic to $\pi_1(M)$, a surface group of genus at least one. Surface groups are Hopfian, so a surjection from one onto an isomorphic copy is an isomorphism. Hence $\pi_1(M) \to \pi_1(U)$ is an isomorphism.
- M and $U \simeq M$ are aspherical, so by Whitehead's theorem the inclusion is a homotopy equivalence.

This uses (a) as well as (a′), and it uses the genus hypothesis: for a sphere the argument fails. Standard facts used without separate checking against a source: the Jordan–Brouwer separation theorem, Tarski–Seidenberg, regular-neighbourhood theory (the regular neighbourhood of a two-sided closed surface is a product), invariance of boundary for topological manifolds, attaching a product collar along a boundary component, that surface groups are Hopfian (they are residually finite), asphericity of closed surfaces of positive genus, Whitehead's theorem, homotopy invariance of the mod-2 degree and its use to decide on which side of an embedded closed surface a point lies, and Alexander duality for the components of the complement of two disjoint closed surfaces. □

4.2. An explicit lower bound for polyhedral surfaces

For $a \in M$, the *conical radius* $\rho(a)$ is the supremum of the $r > 0$ such that $M \cap B(a, r) = C \cap B(a, r)$ for some closed cone C with apex a.

Lemma 4.2 (Cone lemma)**.** *Let $a \in M$, $r < \rho(a)$, and $p \notin M$ with $|p - a| < r/2$. Then every nearest point $x \in \Gamma(p)$ satisfies $\langle p - x, p - a\rangle = |p - x|^2$. In particular p is not critical.*

Proof.

- *Nearest points lie in the ball.* $d(p) \le |p - a| < r/2$, so every $x \in \Gamma(p)$ has $|x - a| < r$.
- *The ray through x stays in M.* The points $y_s = a + s(x - a)$, for s near 1, lie in the cone and in $B(a, r)$, hence in M. So $s \mapsto |p - y_s|^2$ has a local minimum at $s = 1$, which gives $\langle p - x, x - a\rangle = 0$.
- *Conclusion.* $\langle p - x, p - a\rangle = |p - x|^2 + \langle p - x, x - a\rangle = d(p)^2 > 0$ for all $x \in \Gamma(p)$. So 0 is not in the convex hull of the vectors $p - x$.

□

Corollary 4.3. *Let $\lambda(M) = \inf_{x \in M} \sup_{a \in M} \big(\rho(a)/2 - |x - a|\big)$. Then d has no critical point p with $0 < d(p) < \lambda(M)$, so $\operatorname{wfs}(M) \ge \lambda(M)$. If M is a finite union of triangles, $\lambda(M) > 0$.*

Proof.

- *The bound.* Given p with $d(p) < \lambda$ and $x \in \Gamma(p)$, pick a with $\rho(a)/2 - |x - a| > d(p)$. Then $|p - a| < \rho(a)/2$, and Lemma 4.2 applies.
- *Positivity.* For a polyhedron, $\rho(a) > 0$ for every a, by the local conical structure of polyhedra (standard; not separately checked against a source). The function $g(x) = \sup_a(\rho(a)/2 - |x - a|)$ is 1-Lipschitz and finite, since $\rho \le \operatorname{diam} M$, and $g(x) \ge \rho(x)/2 > 0$. M is compact, so $\min g > 0$.

□

Lemma 4.2 and the pointwise form of Corollary 4.3 are machine-checked in Lean 4 with Mathlib, for subsets of an arbitrary real inner product space (Appendix Appendix D).

4.3. A certified lower bound for wfs

For a triangulated reference surface M_{PL} we compute L with "no critical value in $(c \cdot \mathrm{tol}, L)$" by a best-first octree. We call L *certified* in this qualified sense only: the computation is in double precision; containment of a point in a feature is decided with a tolerance $\mathrm{tol} = 10^{-10}$ times the model size; and the search is restricted to distances below a cut-off R.

- *Boxes near M* are certified by Lemma 4.2, using a lower bound for $\rho(a)$ at a candidate point a. The bound comes from the distance from a to the nearest feature (planar region, boundary segment, vertex) that does not contain a. It is sound for planar regions whose directed boundary segments cover their boundary, after a global coplanarity re-check of each merged region.
- *Boxes farther away* are certified by a direction test: a small linear program finds a unit vector u with $\langle u, p - x\rangle > 0$ for all candidate nearest points x of all points p of the box. The candidates are the projections of the box centre onto the relevant features.

- *Stopping rule.* The search stops at the first box that is too small to split further and cannot be certified. Its distance from M is the certified bound L.

Table 1 lists the results. Lemma 4.1 needs no critical value in $(0, r]$. The computed bound covers $(c \cdot \mathrm{tol}, L)$; the interval $(0, c \cdot \mathrm{tol}]$ is covered by Corollary 4.3 whenever $\lambda(M) > c \cdot \mathrm{tol}$, which we have not checked separately.

Part	exact wfs (smooth part)	certified lower bound for M_{PL}
cube, side 10	5	4.530
plate $4 \times 4 \times 1$	0.5	0.436
plate $4 \times 4 \times 0.3$	0.15	0.137
L-shape (wall 2)	1	0.894
L-shape, inner fillet 0.2	1	0.894
block with slot of width 0.2	0.1	0.086
cylinder $R = 3$, $H = 10$	3	2.512

Table 1: Lower bounds for wfs of the triangulated parts, certified in the qualified sense above. The exact column is the wfs of the smooth part; for a triangulated cylinder $\mathrm{wfs}(M_{PL})$ is slightly below 3. The inner fillet caps the reach at 0.2 but leaves the bound unchanged.

5. Existence

We give two constructions that turn hypotheses on the data into a realization. Neither assumes that a perturbed surface embeds as a whole. Theorem 5.1 does assume (P0) that each original patch is embedded, and two of its hypotheses are non-local. Propositions 5.3 and 5.4 assume that the re-trimmed model is embedded, which is checked computationally on the model itself.

5.1. Fans and cones

Setting

K is regular: every face is a disk bounded by a single wire, and every edge has two distinct vertices. The theorem and its proof (Appendix Appendix B) cover only this case. For B-rep complexes, natural readings of the hypotheses exist: at a seam edge E3′ refers to the two sheets of the face; at a closed edge V1 counts edge ends; for a face with inner wires, P0 asks that Q_f be homeomorphic to the face. But the proof does not cover them. For example, at a seam the two fan sheets of one face glue to an annulus. These cases are not proved here. $\mathrm{dev}_e^*(t)$ is the deviation of the boundary points $b_{f,e}(t) = \partial P_f \cap \Pi(t)$ from $\gamma_e(t)$ in the normal plane $\Pi(t)$; it is at most dev_e.

Auxiliary sets

- *Normal disks.* For an edge e, choose parameters $0 < a_e < b_e < 1$ and a C^1 radius function $\rho_e(t)$. ρ_e is constant on the stubs $[t_v, a_e]$, $[b_e, t_w]$ and on a neighbourhood of a_e and b_e; here t_v is the last parameter with $\gamma_e(t) \in B(p_v, r_v)$. $D_e(t)$ is the closed disk of radius $\rho_e(t)$ in $\Pi(t)$ centred at $\gamma_e(t)$, and $T_e = \bigcup_{t \in [a_e, b_e]} D_e(t)$.

- *Vertex regions.* For a vertex v, choose r_v. The vertex region W_v is the union of the ball $B(p_v, r_v)$ and, for each edge e at v, the disks $D_e(t)$ for t between t_v and a_e (or b_e).

Hypotheses

E1–E3′ are required on $[t_v, t_w]$, the stubs included.

E1 $\operatorname{dev}_e^*(t) < \rho_e(t) < \min(\tau_e, \operatorname{reach} \gamma_e|_{[a_e,b_e]})$.

E2 For every face $f > e$:

1. $\psi_f < \pi/2$, where ψ_f is the largest angle between γ_e' and TP_f inside the tube; and $\rho_e k_f < 1$, where $k_f = \kappa_f / \cos\psi_f$;
2. $t \mapsto \pi_e(\beta_{f,e}(t))$ is strictly monotone. A sufficient condition is $L_e + \kappa_\gamma \operatorname{dev}_e^* < 1$, where L_e is the arc-length Lipschitz constant of $\beta_{f,e} - \gamma_e$;
3. *no return:* $\operatorname{dist}(\gamma_e(t), P_f \setminus C_f) > \rho_e(t)$. Here C_f is the union of the section arcs of P_f from $b_{f,e}$ to their exit points, of length at most $\ell_f = (\operatorname{dev}^* + \sqrt{\operatorname{dev}^{*2} + \mu\rho^2})/\mu$ with $\mu = 1 - \rho k_f$.

E3′ The exit points $k_{f,e}(t)$ and $k_{g,e}(t)$ of the two face sections on $\partial D_e(t)$ are distinct. A sufficient condition (E3) is $\alpha_e(t) > \varepsilon_f + \varepsilon_g$, with $\varepsilon = \arcsin\min\{1, (\operatorname{dev}_e^* + k\ell^2/2)/\rho_e\}$. Here α_e is the angle between the inward tangents of the two sections.

V0 γ_e's end at v and the corners of the patches at v lie in $\operatorname{int} B(p_v, r_v)$.

V1 (a) $W_v \subseteq \operatorname{int} \mathcal{Z}(v)$, and W_v is *strictly star-shaped* with respect to p_v: for every $x \in \partial W_v$, the vector $p_v - x$ lies in the interior of the tangent cone of W_v at x, and each ray from p_v meets ∂W_v exactly once. (The second property does not follow from the first at junction points, where a ray could graze ∂W_v, so it is stated separately.) (b) *(non-local)* The construction outside W_v meets ∂W_v in one simple closed curve λ_v.

- Each incident edge contributes one point and each incident face one arc, in the cyclic order of the link of v.
- The outside pieces cross ∂W_v transversally: along the smooth strata of ∂W_v (the ball and the stub cylinders), and along the junction curves where a stub cylinder meets the ball sphere.
- λ_v meets these junction curves transversally and avoids the corners of ∂W_v.

S1 (a) The W_v are pairwise disjoint. (b) The T_e are pairwise disjoint. (c) γ_e and T_e miss W_w for $w \notin \partial e$. (d) P_f misses T_e for $e \not< f$, and misses W_v for $v \not< f$. (e) Outside $\bigcup T \cup \bigcup W$, the patch pieces are pairwise disjoint. (f) $T_e \cap W_v = D_e(a_e)$ for $v < e$, and each piece outside W_v meets W_v only in λ_v.

P0 *(non-local)* Each P_f is embedded, and $P_f \cap (\bigcup T \cup \bigcup W)$ is a collar of ∂P_f, so that $Q_f = P_f \setminus \operatorname{int}(\bigcup T \cup \bigcup W)$ is a closed disk.

V1(b) and P0 are not consequences of local data. In the models of Section 7 they were verified only for the gapped cube, implicitly, through an exact check that the constructed surface is embedded. The sampled checker does not test V1(b), P0, E2.3 beyond S1(d), or S1(e) for adjacent faces.

Theorem 5.1 (Existence by fans and cones; conditional)**.** *Under E1–E3′, V0, V1, S1 and P0, the tolerant B-rep has a realization.*

The proof is in Appendix Appendix B. In outline:

- h follows γ_e along the edges and puts fans of radii in the normal disks.
- The patches, with their collars removed, are kept away from the tubes.
- Cones over the curves λ_v fill the vertex regions.
- Injectivity comes from the disjoint normal disks, E3′, star-shapedness and S1/P0.
- The resulting map is bi-Lipschitz, so a fine secant approximation is a PL realization.

Example 5.2. On a cube whose faces are pulled apart by gaps, the construction was carried out exactly in rational arithmetic: 216 triangles, with zone membership and embeddedness checked exactly (`compute/lemma0_box.py`).

5.2. Re-intersection

Proposition 5.3 (Planar, degree three)**.** *Suppose every face is a disk bounded by a single wire and is planar, $P_f \subseteq \Pi_f$, and every vertex lies on exactly three faces whose planes are independent. Let c_v be the common point of the three planes, $\ell_e = [c_v, c_w]$ for an edge $e = vw$, and $\tilde{P}_f$ the polygon on the c's of the vertices of f, in the order of ∂f. Assume:*

- *$c_v \in \mathcal{Z}(v)$, $\ell_e \subseteq \mathcal{Z}(e)$ and $\tilde{P}_f \subseteq \mathcal{Z}(f)$;*
- *the $\tilde{P}_f$ are simple polygons that meet exactly in their common edges and vertices.*

Then the cellular map onto $\bigcup \tilde{P}_f$ is a realization.

Proof. It is a PL homeomorphism onto $\bigcup \tilde{P}_f$, and each cell lands in its zone. □

Proposition 5.4 (Curved re-intersection)**.** *Suppose there is a* re-trimmed model*: compact patches $\tilde{P}_f$, curves $\tilde{\gamma}_e$ and points c_v, with $\Sigma = \bigcup_f \tilde{P}_f$, satisfying the following.*

P *Each face has a C^2 regular parametrization $X_f \colon U_f \to \mathbb{R}^3$, where U_f is open in $\mathbb{R}^2$ or in a quotient $\mathbb{R}^2/\Lambda_f$ by a lattice of periods (this is how seams are represented). There is a compact region $D_f \subset U_f$ with $X_f(D_f) = \tilde{P}_f$ and X_f injective on D_f. Singular parametrizations, such as sphere poles and cone apices, are excluded.*

C *There is a homeomorphism $\phi \colon |K| \to \Sigma$ mapping each open cell of K onto the corresponding open cell of Σ. The open cells of Σ are the points c_v, the curves $\tilde{\gamma}_e$ minus their end points, and the patches minus all their edge curves (seams included).*

R *Let $G_f \subset D_f$ be the* edge graph *of f: the preimage under X_f of all edge curves of f, including seams, which in the quotient parametrization lie inside D_f rather than on its boundary. G_f is a finite graph. Its vertices are the preimages of the points c_v. Its edges are simple C^2 regular arcs, or loops at a vertex, meeting only at vertices. At each vertex of G_f, every sector of D_f between two consecutive arcs has angle in $(0, 2\pi)$. A smooth passage, with angle π, is allowed: for example a seam vertex on a smooth boundary curve.*

B1–B3 *$c_v \in \mathcal{Z}(v)$, $\tilde{\gamma}_e \subseteq \mathcal{Z}(e)$ and $\tilde{P}_f \subseteq \mathcal{Z}(f)$.*

T *At every interior point of every edge curve $\tilde{\gamma}_e$, the inward tangent half-planes of the two adjacent patches (or of the two sides of a seam) are distinct: the surface does not fold back on itself. At every vertex c_v, the tangent cone of Σ is an embedded disk. That cone is the union of the tangent sectors $dX_f(\text{sector})$, taken over all sectors of the edge graphs G_f at preimages of c_v. The condition means these sectors meet only along common boundary rays, in the cyclic order of the link of v. (Together with the C^1 regularity up to the vertices, this also controls the edge condition at the end points of the edges.)*

Then the tolerant B-rep has a realization.

Remark 5.5.

- By C, Σ is a disjoint union of closed surfaces, so it bounds a solid in the sense of Definition 2.5.
- C covers seams, closed edges and faces with inner wires.
- The edge part of T asks only that the surface does not fold. Transversal faces satisfy it, and so do tangent (G^1) edges, where the two half-planes are opposite. So the proposition applies in principle to filleted models, provided a re-trimmed model with the stated properties is available. Our computation builds the re-trimmed model by re-intersecting extended surfaces, which fails for tangent faces (Section 7); that is a limitation of the computation, not of the proposition.

The proof is in Appendix Appendix C.

Remark 5.6 (Checking the hypotheses).

- The re-trimmed model can be computed by extending the face surfaces and rebuilding the solid (OpenCascade's `BOPAlgo_MakerVolume`). The hypotheses are then checked on samples (`compute/reintersect_curved.py`), except the vertex part of hypothesis T, which is not checked. That matters for the cylinder and the pipe, whose seam vertices lie on only two surfaces. The PL step of the proof is not carried out numerically; it is an existence argument.
- For the planar case, the hypotheses of Proposition 5.3 for the sewn box and the sewn plate were also re-checked in exact rational arithmetic, including embeddedness.
- Results are in Section 7.

6. An obstruction

6.1. Setting

- $f \neq g$ are faces whose closed cells are disks bounded by single wires.
- Their data are planar: $P_f \subseteq \Pi_f$ and $P_g \subseteq \Pi_g$, with the planes not parallel.
- $\lambda = \Pi_f \cap \Pi_g$, and $\alpha \in (0, \pi/2]$ is the angle between the planes.
- n_f, n_g are unit normals, and m_f is a unit vector in Π_f orthogonal to λ (likewise m_g).

- π_f is the orthogonal projection onto Π_f, and $U_f = \bigcup_{e<f} \mathcal{Z}(e)$, which contains the zones of the vertices of f.
- ℓ_f is the boundary loop of f: the arcs $\beta_{f,e}$ in the cyclic order of ∂f, with consecutive arcs joined by the segment between their end points.

Theorem 6.1 (Obstruction)**.** *Suppose there are $X \in \lambda$ and $\rho > 0$ such that, with $J_f = X + [-\rho, \rho]m_f$ and $J_g = X + [-\rho, \rho]m_g$:*

O0 *$\beta_{f,e} \subseteq \mathcal{Z}(e)$ for every edge e of f, and at every vertex v of f the end points of the two arcs of f at v lie in $\mathcal{Z}(v)$; likewise for g.*

O1 *$J_f \cap \ell_f = \emptyset$, and the winding number of ℓ_f in Π_f about the points of J_f is odd; likewise for g.*

O2 *$J_f \cap \operatorname{cl} \pi_f(U_f) = \emptyset$ and $J_g \cap \operatorname{cl} \pi_g(U_g) = \emptyset$.*

O3 *$\mathcal{Z}(e)$ is simply connected for every edge e of f or g.*

O4 *$\rho \sin\alpha > \tau_f \cos\alpha + \tau_g$ and $\rho \sin\alpha > \tau_g \cos\alpha + \tau_f$.*

Then the tolerant B-rep has no realization.

Proof. Suppose h is a realization, and let $\varphi = \pi_f \circ h|_{\bar{f}}$.

Step 1: over J_f, $h(\bar{f})$ lies in the slab.

- Let $y \in J_f$. We have $h(\bar{f}) \subseteq \mathcal{Z}(f) = N_{\tau_f}(P_f) \cup U_f$, and by O2 no point of U_f lies over y.
- Since $P_f \subseteq \Pi_f$, $\pi_f^{-1}(y) \cap N_{\tau_f}(P_f) \subseteq \sigma_y := y + (-\tau_f, \tau_f)n_f$.
- So $h(\bar{f}) \cap \pi_f^{-1}(y) = h(\bar{f}) \cap \sigma_y$. These points are images of interior points of f, because $h(\partial f) \subseteq U_f$.

Step 2: degree one.

- Parametrize ℓ_f over ∂f: a small arc A_v around each vertex goes to the joining segment, and the rest of each edge goes to $\beta_{f,e}$. Shrink A_v until $h(A_v) \subseteq \mathcal{Z}(v)$.
- Build a homotopy in U_f from $h|_{\partial f}$ to this parametrization:
  - on A_v, the straight-line homotopy inside the convex ball $\mathcal{Z}(v)$;
  - on the rest of an edge e, the boundary of the square is mapped into $\mathcal{Z}(e)$ (by O0, monotonicity, and $h(\bar{e}) \subseteq \mathcal{Z}(e)$), so it extends over the square by O3.
- By O2 the projected homotopy avoids every $y \in J_f$. So $\varphi|_{\partial f}$ and ℓ_f have the same winding number about y, which is odd by O1. Hence the mod-2 degree of φ at y is 1.
- Fix a triangulation of $\bar{f}$ on which h is linear. For $y \notin \varphi(1\text{-skeleton})$, which excludes only finitely many points of J_f once the slice in Step 3 is generic, $\#(h(\bar{f}) \cap \sigma_y)$ is odd.

Step 3: a crossing arc.

- Let Σ be the plane through X orthogonal to λ; it contains n_f, m_f, n_g, m_g.
- Move X along λ and shrink ρ slightly to generic values; O1, O2 and O4 are open conditions. Genericity means: Σ contains no vertex of the triangulations and meets their edges transversally, and the lines $\langle x - X, m_f\rangle = \pm\rho$, $\langle x - X, m_g\rangle = \pm\rho$ contain no vertex.
- In Σ, the rectangle $R_f = J_f + [-\tau_f, \tau_f]n_f$ meets $h(\bar{f})$ in a compact 1-manifold M_f.
- M_f misses $h(\partial f)$, since $\pi_f(R_f) = J_f$ and O2 holds. It misses the long sides of R_f, by Step 1. So its boundary lies on the two short sides.
- A generic fibre σ_y meets M_f an odd number of times (Step 2). Circle components, and arcs with both ends on one short side, meet it an even number of times.
- Hence some arc $a_f \subseteq M_f$ joins the two short sides of R_f. Likewise there is an arc a_g for g.

Step 4: the arcs meet.

- In Σ, use the coordinates $u = \langle x - X, n_f\rangle$ and $w = \langle x - X, n_g\rangle$.
- On the short sides of R_f, $|w| \geq \rho \sin\alpha - \tau_f \cos\alpha > \tau_g$ by O4, with opposite signs on the two sides. So a_f contains a sub-arc in the parallelogram $Q = \{|u| \leq \tau_f, |w| \leq \tau_g\}$ joining $w = -\tau_g$ to $w = \tau_g$. Symmetrically, a_g contains a sub-arc in Q joining $u = -\tau_f$ to $u = \tau_f$.
- By O4, Q lies in both rectangles.
- Two paths in a square joining the two pairs of opposite sides intersect (a standard consequence of the Brouwer fixed point theorem; not separately checked against a source). So $h(\bar{f}) \cap h(\bar{g}) \neq \emptyset$ at a point that is the image of interior points of both f and g (Step 1). This contradicts injectivity.

□

Remark 6.2.

- *Which hypotheses bind.* Face tolerances are tiny (OpenCascade's default is 10^{-7}), so O4 is mild. The binding conditions are O1 and O2: the patches must cross away from the shadows of the edge and vertex zones.
- *Curved patches.* The proof uses planarity in Steps 1 and 2. A version for C^2 patches with τ_f below their reach should follow with tubular coordinates; it is not written.
- *Standard facts used without separate checking against a source:* mod-2 degree of PL maps and its relation to the winding number, general position for PL maps and generic planes, parity of crossings of a segment with a PL circle, homotopy invariance of the winding number, the nerve lemma (for the O3 test in the computations), and the square lemma cited above.

7. Experiments with OpenCascade

7.1. Set-up

- *Software.* All experiments use OpenCascade 8.0.1 through `cadquery-ocp`. Scripts and results are in the repository [24].
- *The two checks.* "BRepCheck" is `BRepCheck_Analyzer(shape).IsValid()`. "BOP check" is `BRepAlgoAPI_Check(shape, true, true)`, which also tests geometry and self-interference.
- *Sewn models* imitate imported data. Every face of an exact part is moved by a random vector of length between $g/2$ and g, and the faces are sewn back with `BRepBuilderAPI_Sewing` at tolerance $4g$. Sewing sets each edge tolerance to 1.05 times its own deviation estimate, which is 1.048–1.05 times our measured deviation.
- *Scope.* These are a few parts in one kernel version, not a survey of industrial models.

7.2. The checks do not decide existence or uniqueness

Table 2 summarises.

Model	BRepCheck	BOP check	What is true
twisted triangular prism, two faces with linked boundaries ($\tau = 0.081$)	valid	self-intersection	no realization (linking number -1)
thick pipe, tolerance 1.3× hole radius	valid	valid	two realizations with non-homeomorphic solids (knot determinant 3 vs. 1)
sewn wedges, 30° and 10°, $g = 0.01$	valid	valid	no realization (Theorem 6.1)
sewn plate, thickness 0.05, $g = 0.01$	valid	invalid	realizable (Proposition 5.3)
filleted L-shape, $\tau = 0.1$	valid	self-intersection	$\tau + \text{defl} <$ certified wfs bound of M_{PL}
long cylinder, $\tau = 3.3$	valid	valid	$\tau \geq$ wfs(nominal) (critical point exhibited)

Table 2: OpenCascade's checks against what the theory shows.

Twisted prism

- Two non-adjacent triangular faces A, B have boundary loops with linking number -1 (by crossing count; a numerical Gauss integral agrees). The loops are $0.4037 > 2\tau = 0.1615$ apart, so the unions T_A, T_B of their edge and vertex zones are disjoint.
- In a realization, each $h(e)$ lies in the convex capsule $C_e = N_\tau(e)$, which contains the end balls (all tolerances are equal here). Homotope $h(e)$ rel end points to a segment inside C_e, and the vertices to the nominal ones inside the balls. This gives a homotopy of $h(\partial A)$ to ∂A inside T_A, so in the complement of T_B; likewise for B.

- The linking number is invariant under such homotopies, so $\mathrm{lk}(h(\partial A), h(\partial B)) = -1$, and $h(\partial A)$ must meet the disk $h(\bar{B})$. That contradicts embeddedness.

Knotted pipe

- One realization keeps the nominal geometry. The other replaces the inner face by a thin tube, of radius 0.005, around a centreline that follows a long trefoil.

- The script checks the zones (maximum distance $0.983 \leq 1.3$), embeddedness ($\varepsilon\kappa_{\max} = 0.31$; non-local distance $0.036 > 2\varepsilon$ outside a window of $0.04 < \pi/\kappa_{\max} = 0.0504$) and the knot (determinant 3).

- The second solid is the exterior of a knotted arc in a ball, whose fundamental group is non-abelian: a knot of determinant 3 admits a non-trivial Fox 3-colouring, and a non-trivial 3-colouring of a diagram defines, through the Wirtinger presentation, a surjection of the knot group onto the non-abelian group S_3 (standard; not separately checked against a source). The nominal solid is a solid torus. So the two solids are not homeomorphic.

- This shows that the collar hypothesis of Theorem 3.5 cannot be dropped. Here $\mathrm{wfs} \leq 1 < 1.3$.

Sewn wedges

- At the knife edge of the sewn wedges, the two planar patches cross along a line that lies outside the edge tube, by 49% (30°) and about 3× (10°) of τ_e.

- The hypotheses of Theorem 6.1 hold. For the 30° wedge the largest feasible ρ at the positions searched is $1.6 \cdot 10^{-3}$, while O4 needs only $3.7 \cdot 10^{-7}$; for the 10° wedge it is $1.6 \cdot 10^{-2}$.

- An independent check in exact rational arithmetic confirmed all of O0–O4 for both wedges.

- On the sewn box and plate and on the exact box and wedge, the checker finds no crossing witness, consistent with their being realizable.

7.3. Which construction applies

- *Why the fans fail on sewn data.* The fan construction fails at the vertices of sewn models. Sewing sets $\tau_e = 1.05\,\mathrm{dev}_e$, and the tube radius must exceed the deviation; but at a corner of a box three tubes of radius greater than dev do not fit disjointly into a vertex ball of radius about τ_e.

- *Re-intersection.* Re-intersection avoids this. Our computation of the re-trimmed model, which re-intersects the extended surfaces, needs transversal faces. "Pass (curved)" refers to the edge part of hypothesis T only; the vertex part is not checked.

- *Sewn models with tangent edges remain open.* On the sewn filleted box, OpenCascade sets each vertex tolerance equal to the largest incident edge tolerance, the patch corners lie up to $0.85\,\tau_v$ from the vertex point, and none of our vertex constructions fits.

Model	fans (Thm. 5.1)	re-intersection	margins of re-intersection (vertex / edge / face)
exact box, cylinder, pipe, slot, filleted box, 10° wedge	pass	pass where applicable	—
L-shape, all edges filleted 0.5	pass with variable radius*	not applicable	—
L-shape fused, inner fillet 0.2	fails (2 cusp vertices, 9 V1(a) failures); passes after merging coplanar faces	not applicable	—
sewn box $g = 0.01$	fails at vertices	pass	45% / 13% / 4.9%
sewn plate $g = 0.01$	fails at vertices	pass	45% / 13% / 5.9%
sewn cylinder $g = 0.01$	fails at vertices	pass (curved)	55% / 35% / 5.0%
sewn cylinder $g = 0.05$	fails at vertices	pass (curved)	55% / 11% / 8.9%
sewn pipe $g = 0.01$	fails at vertices	pass (curved)	53% / 16% / 5.1%
sewn wedges	fails at vertices	fails (edge outside tube)	no realization (Thm. 6.1)
sewn filleted box $g = 0.001$	fails at vertices	computation not applicable (tangent faces)	open

Table 3: Constructions on exact and sewn models (sampled checks, double precision). *Provisional: the checker does not yet implement the constant radius near the vertices. Face margins are the smaller of the sampled checker's value and an independent dense re-check with exact projections (`results/reintersect_B3_dense.txt`); for the sewn plate only the dense re-check is available.

7.4. Other observations

- *Thickness heuristics.* Half the minimum distance between non-adjacent faces is a common estimate of the safe tolerance. It overestimates the cylinder (5 against wfs = 3), and underestimates an L-shape with a 0.05 inner fillet by a factor of about 97.

- *Offsets.* In its default mode, `BRepOffsetAPI_MakeOffsetShape` returned solids with wrong volume (errors from 44% to over a thousandfold, with six shells instead of one in every such case) for a filleted box at inward offsets larger than the fillet radius but within the wfs-safe range, and BRepCheck accepted them. A plausible mechanism is in the source of OpenCascade 8.0.1: when the offset radius of a cylindrical, spherical or toroidal fillet face becomes negative, its offset surface gets the status `BRepOffset_Reversed`, and `BRepOffset_MakeOffset::BuildOffsetByArc` skips such faces (source comment: "Degenerated or returned faces are not stored"). We have not traced the rest of the algorithm. These failures appear not to involve tolerances; they look like failures of face-by-face offsetting when faces must disappear.

- *Why BRepCheck ignores the size of tolerances.* In OpenCascade 8.0.1 the documentation comment of `BRepCheck_Analyzer` lists the status `BRepCheck_InvalidToleranceValue`

as "NYI" (not yet implemented) for vertices, edges and faces. In `BRepCheck_Analyzer.cxx` the flag that would set this status on a face is initialised to false and never assigned, and no other file sets the status. So BRepCheck never reports a tolerance as too large. (This was first noticed in a companion book project; we checked the source ourselves.)

- *Public documentation.* We did not find a check that tolerances leave the topology unchanged in the public documentation of OpenCascade, or in third-party documentation of ACIS and Parasolid. That says only that the documentation does not mention one.

8. Limitations and open problems

1. **Citations behind the proofs.** The results quoted from the literature in the proofs have been checked against the originals. What remains are the standard facts listed, each time, as "not separately checked against a source".
2. **B-rep complexes in the fan construction.** Theorem 5.1 is proved for regular complexes only; seams, closed edges and faces with inner wires need separate arguments. (Proposition 5.4 covers them.)
3. **Spheres.** Theorem 3.5 excludes shells of genus 0. Waldhausen's horizontality argument does not apply to them. The PL Schoenflies theorem [14, Thm. 17.12] is the natural tool for the sphere case; the argument is not written.
4. **Embeddings.** We prove that the images of realizations are ambient isotopic, not that the embeddings are isotopic as maps. The two differ by a self-homeomorphism of the surface.
5. **Tight vertices.** We do not know whether sewn models with tangent edges and OpenCascade's vertex tolerances (equal to the largest incident edge tolerance) have realizations. Two directions are open: a vertex construction that uses the edge tubes as well as the ball, or a vertex version of the obstruction.
6. **Re-trimmed models for tangent edges.** Proposition 5.4 allows tangent (G^1) edges and mixed vertices, but our computation of a re-trimmed model re-intersects extended surfaces and fails when faces are tangent. A construction of re-trimmed models for such data is open.
7. **Curved obstruction.** Theorem 6.1 is stated for planar patches.
8. **Rigour of the computations.**
   - The wfs bounds and the hypothesis checks use double precision and sampling. Exceptions: the gapped-cube construction, and the exact re-verification of the sewn wedges, the sewn box and the sewn plate.
   - Interval arithmetic and a compiled implementation are needed for industrial models.
9. **Formalization.** The carrier lemma in general (contractible carriers), Waldhausen's theorem and the existence constructions are not formalized.

Data and code availability

All scripts, computed results and the Lean formalization are available in the public repository [24].

Declaration of competing interest

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the research and writing process

During this work the author used Claude (Anthropic) to draft proofs and text, write and run the experiment scripts and the Lean formalization, and check citations against the original sources; independent AI review passes were used to check proofs and citations. Lean 4 with Mathlib verified the formalized lemma. The author directed the research, reviewed and edited the content, and takes full responsibility for the publication.

Appendix A. A curved-triangle lemma

Appendices Appendix B and Appendix C need maps of triangles with one curved side that reproduce a *prescribed* parametrization of that side. This is what makes maps built separately on two adjacent faces agree on their common edge. We give an explicit construction.

Lemma Appendix A.1. *Let $\hat{c}\colon [t_0, t_1] \to \mathbb{R}^2$ be a C^1 regular curve, $a_1 = \hat{c}(t_0)$, $a_2 = \hat{c}(t_1)$, and a_3 a point.*

- *Put $\psi(s) = \hat{c}(t_0 + s(t_1 - t_0)) - ((1-s)a_1 + sa_2)$ for $s \in [0,1]$.*
- *In barycentric coordinates $(\lambda_1, \lambda_2, \lambda_3)$ of the reference triangle Δ, put $\mu = \lambda_1 + \lambda_2$ and*

$$F(\lambda) = \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 + \mu^2\, \psi(\lambda_2/\mu) \qquad (F = a_3 \text{ at } \mu = 0).$$

Then:

(a) on the side $\lambda_3 = 0$, F is $\lambda_2 \mapsto \hat{c}(t_0 + \lambda_2(t_1 - t_0))$;
(b) on the sides $\lambda_1 = 0$ and $\lambda_2 = 0$, F is affine;
(c) F is C^1 on the closed triangle Δ, including at its corners.

Moreover, consider a family of such triangles in which the chord triangles (a_1, a_2, a_3) are shape-regular with diameter $h \to 0$, the speed $|\hat{c}'|$ is bounded above and below by positive constants (so $|t_1 - t_0| = O(h)$), and $\hat{c}'$ has a uniform modulus of continuity ω, which we take subadditive (replacing it by its concave majorant if necessary). Both hold in Appendices Appendix B and Appendix C, where the curves are fixed regular C^1 curves on compact parameter intervals. Then for small h:

(d) F is a C^1 diffeomorphism of Δ onto the region bounded by the arc $\hat{c}([t_0, t_1])$ and the segments $[a_2, a_3]$ and $[a_3, a_1]$, and $DF = A(I + O(\omega(h)))$, where A is the derivative of the affine part.

Proof. (a) On $\lambda_3 = 0$ we have $\mu = 1$, so $F = (1 - \lambda_2)a_1 + \lambda_2 a_2 + \psi(\lambda_2) = \hat{c}(t_0 + \lambda_2(t_1 - t_0))$.

(b) $\psi(0) = \psi(1) = 0$. On $\lambda_2 = 0$ the correction is $\mu^2\psi(0) = 0$. On $\lambda_1 = 0$ it is $\mu^2\psi(1) = 0$ (for $\mu > 0$).

(c) For $\mu > 0$, F is C^1. Write $G = \mu^2\psi(s)$ with $s = \lambda_2/\mu$. Then

- $|G| \le \mu^2 \max|\psi| = O(\mu^2)$, so G is differentiable at a_3 with $DG(a_3) = 0$;
- $dG = 2\mu\psi(s)\,d\mu + \mu^2\psi'(s)\,ds$ with $|ds| = O(1/\mu)$, so $|dG| = O(\mu) \to 0$ as $\mu \to 0$.

Hence F is C^1 at a_3, with $DF(a_3) = A$. At the other two corners $\mu = 1$, and ψ is C^1 on $[0,1]$.

(d) *Size of the correction.* Put $\Delta t = t_1 - t_0$. Then $\psi'(s) = \Delta t\,\big(\hat{c}'(t_0 + s\Delta t) - \overline{\hat{c}'}\big)$, where $\overline{\hat{c}'}$ is the mean of $\hat{c}'$ over $[t_0, t_1]$. So $|\psi'| \le |\Delta t|\,\omega(|\Delta t|) = O(h\,\omega(h))$, and since $\psi(0) = 0$, also $|\psi| = O(h\,\omega(h))$. Hence $|DG| = O(h\,\omega(h))$ on Δ, in coordinates of the unit reference triangle.

Injectivity. By shape regularity $\|A^{-1}\| = O(1/h)$, so $DF = A + DG = A(I + O(\omega(h)))$. For $x, y \in \Delta$ (which is convex),

$$|F(x) - F(y)| \ge |A(x-y)| - Ch\,\omega(h)|x-y| \ge (ch - Ch\,\omega(h))|x-y| > 0$$

for small h. So F is injective, and its differential is invertible.

Image. F maps $\partial\Delta$ homeomorphically onto the closed curve formed by the arc and the two segments. That curve is simple for small h: the arc stays within $O(h\,\omega(h))$ of its chord, and the two segments make angles with the chord that are bounded below by shape regularity. An injective continuous map of a closed disk whose boundary goes onto a Jordan curve has as image the closed region bounded by that curve. □

Remark. The classical blending map with $\mu\,\psi(\lambda_2/\mu)$ in place of $\mu^2\psi(\lambda_2/\mu)$ is exact on the curved side and affine on the straight sides. But it is only Lipschitz at a_3: its derivative there depends on the direction of approach. The factor μ^2 is what gives (c).

Standard facts used without separate checking against a source: the Jordan curve theorem, and that an injective continuous map of a closed disk onto a set bounded by its image boundary curve has the closed Jordan region as its image (invariance of domain).

Appendix B. Proof of Theorem 5.1

We use the notation and hypotheses of Section 5.1; K is regular.

Exit points. For a face $f > e$, let $c(\sigma)$ be the section of P_f in $\Pi(t)$, parametrized by arc length from $b_{f,e}(t)$ and written relative to $\gamma_e(t)$. Then $(|c|^2/2)'' = 1 + \langle c, c''\rangle \ge 1 - \rho_e k_f > 0$ by E2.1. So $|c|$ cannot touch the circle $\partial D_e(t)$ from inside before crossing it, and it crosses with $\langle c', \hat{r}\rangle > 0$, where $\hat{r}$ is the outward radial direction. The *exit point* $k_{f,e}(t)$ is that crossing point. It is unique by E2, and C^1 in t by the implicit function theorem.

Margin. $m > 0$ is the minimum zone margin. $\rho_e < \tau_e$ and $W_v \subseteq \operatorname{int}\mathcal{Z}(v)$, and by compactness all pieces built below lie at distance at least m inside the zones of their cells.

Arcs. For a face f at v, we use two arcs:

- $A_{f,v}$ is the arc of ∂Q_f on ∂W_v, from $k_{f,e}(a_e)$ to $k_{f,e'}(a_{e'})$;
- $\Lambda_{f,v}$ is the arc of λ_v contributed by f, from $\gamma_e(a_e)$ to $\gamma_{e'}(a_{e'})$. It consists of the two radii $[\gamma_e(a_e), k_{f,e}(a_e)]$ in the end disks, followed by $A_{f,v}$.

Appendix B.1. The image pieces

1. *Vertices.* $h(v) = p_v$.
2. *Edges.* On $[a_e, b_e]$, $h(e) = \gamma_e$.
3. *Fans.* The fan strip is $F_{f,e} = \{\gamma_e(t) + \sigma(k_{f,e}(t) - \gamma_e(t)) : t \in [a_e, b_e],\ \sigma \in [0,1]\}$.
4. *Faces away from the vertices.* $Q_f = P_f \setminus \operatorname{int}(\bigcup T \cup \bigcup W)$.
5. *Vertex regions.* Inside W_v, take the cone from p_v over λ_v.
   - The cone over the point $\gamma_e(a_e)$ is the stub of $h(e)$.
   - The cone over $\Lambda_{f,v}$ is the sector of $h(f)$ at v.

Appendix B.2. Corners of λ_v

λ_v is piecewise C^1. Its corners are:

- the points $\gamma_e(a_e)$, where two fan radii meet;
- the points $k_{f,e}(a_e)$, where a radius in the end disk meets the trace of Q_f;
- the points where λ_v crosses a junction curve of ∂W_v. By V1(b) these crossings are transversal and avoid the corners of ∂W_v.

Between consecutive corners, λ_v is a C^1 curve with nonvanishing tangent.

Appendix B.3. Injectivity of the image

Inside T_e.

- The disks $D_e(t)$ lie in the normal planes and have radius below the reach of $A = \gamma_e([a_e, b_e])$, so they are pairwise disjoint. Indeed, by Federer's Theorem 4.8(2),(12) [10, pp. 434–435], if $a \in A$, v lies in the normal cone $\operatorname{Nor}(A, a)$ and $|v| < \operatorname{reach}(A)$, then a is the unique nearest point of A to $a + v$. The tangent cone is the tangent line at interior points and a ray at the end points; its dual, the normal cone, is then the normal plane, respectively a half-space, and contains the normal plane in both cases. So every point of $D_e(t)$ has nearest point $\gamma_e(t)$. This holds whatever $\rho_e(t)$ is.
- Inside one disk, the construction is the centre and two radii to the distinct points $k_{f,e}(t)$ and $k_{g,e}(t)$ (E3′). These meet only at the centre.
- No other piece enters T_e:
  - non-incident pieces are excluded by S1(b)–(d) and S1(f);
  - the patch of $f > e$ meets $D_e(t)$ only in the section arc from $b_{f,e}(t)$ to $k_{f,e}(t)$ (by the convexity computation above and E2.3). That arc lies in the collar of ∂P_f, which is removed (P0).

Inside W_v.

- The cone map $(s, x) \mapsto p_v + s(x - p_v)$, $s \in (0, 1]$, $x \in \lambda_v$, is injective. Each ray from p_v meets ∂W_v once (V1(a)), and λ_v is simple (V1(b)).

- No other piece enters W_v: S1(a), (c), (d) exclude non-incident pieces, and S1(f) excludes incident ones.

Elsewhere.

- Outside the closed set $\bigcup T \cup \bigcup W$, the pieces are the embedded Q_f (P0), and they are pairwise disjoint (S1(e)).
- On ∂T_e, two different Q_f could meet only at exit points, which are distinct (E3′), or at non-incident patches, which are excluded by S1(d).
- On ∂W_v, they meet λ_v in disjoint arcs, because λ_v is simple.

Cell structure.

- By P0, ∂Q_f is one circle made of the curves $k_{f,e}$ and the arcs $A_{f,v}$.
- By V1(b) there is exactly one arc per corner of f. So the pieces form a 2-regular graph isomorphic to ∂f; its only cycle is ∂Q_f, and the cyclic order is forced.
- The pieces meet exactly as follows:
  - $Q_f \cap F_{f,e} = k_{f,e}$, by E2.3 and S1(f);
  - $(Q_f \cup \text{fans}) \cap \text{sector}_{f,v} = \Lambda_{f,v}$, by strict star-shapedness and S1(f).
- Gluing two disks along a boundary arc that is their whole intersection gives a disk. By induction, the union $h(f)$ is a closed disk with $\partial h(f) = \bigcup_{e<f} h(e)$.

Zones.

- $Q_f \subseteq P_f \subseteq \mathcal{Z}(f)$.
- $F_{f,e} \subseteq T_e \subseteq N_\rho(\gamma_e) \subseteq \mathcal{Z}(e) \subseteq \mathcal{Z}(f)$.
- The cone parts lie in $W_v \subseteq \mathcal{Z}(v)$, which is contained in $\mathcal{Z}(e)$ and $\mathcal{Z}(f)$ for incident e and f.
- $h(\text{int}\, e) \subseteq \gamma_e \cup W_v \subseteq \mathcal{Z}(e)$.
- All inclusions hold with margin at least m.

Appendix B.4. The map h

We build a polyhedral domain and a homeomorphism onto the union of the pieces that is C^1, with injective differential, on each closed domain cell (except at the apexes), and that agrees on shared edges.

Domain. Let L be a polyhedral subdivision of $|K|$ with 2-cells for the fans, for the pieces Q_f and for the sectors, in the incidence pattern of the image pieces. Each sector is subdivided into *sub-cones*: cones from the vertex v over one segment of a polygonal arc of the link of v. That arc has a vertex at the preimage of every corner of λ_v, so the image of a sub-cone is the cone over a C^1 piece of λ_v.

The parametrization ϕ. Fix once and for all a parametrization ϕ of λ_v by the polygonal link, C^1 with $\phi' \neq 0$ on each link segment, such that:

- on each end radius $[\gamma_e(a_e), k_{f,e}(a_e)]$ it is affine in the fan parameter σ;
- on each arc $A_{f,v}$ it is some fixed C^1 regular parametrization.

The maps below all use this ϕ.

Fans. Use the chart $(t, \sigma) \mapsto \gamma_e(t) + \sigma(k_{f,e}(t) - \gamma_e(t))$ on $[a_e, b_e] \times [0, 1]$, precomposed with a bilinear map from the quadrilateral domain cell onto this rectangle.

- It is C^1 (since $k_{f,e}$ is).
- Its differential is injective: $\partial_\sigma = k - \gamma \in \Pi(t)$ is nonzero, and ∂_t has γ'-component at least $1 - \sigma\kappa_\gamma\rho_e > 0$.
- It fixes the parametrization of the crease $k_{f,e}$ by t, and that of the end radius at $t = a_e$ by σ.

Sectors. On each sub-cone, map $v + s(y - v)$ to $p_v + s(\phi(y) - p_v)$. This map is C^1 up to the boundary away from v, with injective differential, because $\phi' \neq 0$ and ϕ' is transverse to the rays (V1(a)). It agrees with the fan on the end radii (both affine in σ), and with the map on Q_f along $A_{f,v}$ (both use ϕ).

Q_f. In the parameter domain of P_f, ∂Q_f is a piecewise C^1 curve. Its pieces are $k_{f,e}$, parametrized by t, and $A_{f,v}$, parametrized by ϕ. Its corners are at the points $k_{f,e}(a_e)$ and at the junction crossings.

- Subdivide Q_f into curved triangles, each with at most one curved edge. The curved edges are pieces of ∂Q_f with their prescribed parametrizations, and the corners of ∂Q_f are vertices. Triangulate the domain cell of Q_f in L to match.
- On each curved triangle, use the map of Lemma Appendix A.1. It reproduces the prescribed parametrization on the curved edge, is affine on the straight edges, is C^1 up to the corners, and is a diffeomorphism for a fine enough subdivision. As in Appendix Appendix C, put one interior vertex near each corner of ∂Q_f so that no triangle has two curved edges.
- Composed with the chart of P_f, this gives maps that are C^1 up to the boundary, have injective differential, and agree on shared edges.

These cell maps agree on shared edges, so together they define a homeomorphism $h\colon |L| = |K| \to \bigcup$ pieces. It sends each cell of K onto the corresponding image cell.

Appendix B.5. The PL step

Metric. We measure Lipschitz constants on each shell of $|L|$ in its intrinsic length metric. That metric is bi-Lipschitz equivalent to the extrinsic one, and in it a Lipschitz bound on each cell is a global bound. Injectivity between shells is argued separately at the end.

h is bi-Lipschitz. h is injective on a compact space, so it suffices that it is locally bi-Lipschitz.

- *Inside a cell*, this follows from the injective differential.

- *Along the creases*, the two half-planes of the tangent cone must differ.
  - Along k (fan and Q_f): in $\Pi(t)$ the fan leaves k in direction $-\hat{r}$ and the patch section in direction c' with $\langle c', \hat{r}\rangle > 0$. Since $k' \notin \Pi(t)$, the two half-planes could coincide only if c' were parallel to $-\hat{r}$, which is excluded.
  - Along γ_e (two fans): E3′ and compactness.
  - Along ∂W_v (cone and outside piece): the cone's half-tangent $p_v - x$ points strictly into W_v (strict star-shapedness, V1(a)), and the outside piece leaves strictly outward (transversality, V1(b)). For a fan, the tangent plane along the end radius contains $\partial_t = (1-\sigma)\gamma' + \sigma k'$, which does not lie in $\Pi(a_e)$.
  - Along the rays from p_v to a corner of λ_v (between two sub-cones): by homogeneity, the opening angle along the ray is constant away from p_v, and it is positive because the radial projection of λ_v to the unit sphere about p_v has no fold at the corner (see the corner arguments below and V1(a)).
- *At $k(a_e)$* (fan, Q_f and two cone sub-sectors). Here $\rho_e'(a_e) = 0$, so $k' \perp \hat{r}$. Let $Q = \{\langle \cdot, \gamma'\rangle \le 0, \langle \cdot, \hat{r}\rangle \le 0\}$ and $w = p_v - k(a_e)$; then $w \in \operatorname{int} Q$ by strict star-shapedness.
  - The fan sector $\operatorname{span}^+(k', -\hat{r})$ lies in $\{\langle \cdot, \gamma'\rangle \ge 0\}$.
  - The Q_f half-plane, bounded by $\mathbb{R}k'$ and containing c', lies in $\{\langle \cdot, \hat{r}\rangle \ge 0\}$.
  - The two cone sub-sectors $\operatorname{span}^+(-\hat{r}, w)$ and $\operatorname{span}^+(w, -k')$ lie in Q.

  So the fan and Q_f sectors lie in the closure of the complement of Q, and the cone sectors lie in Q. They share only the rays k' (fan and Q_f), $-\hat{r}$ (fan and cone), w (between the two cone sub-sectors) and $-k'$ (cone and Q_f). The two cone sub-sectors do not fold: if $-\hat{r}$, w, $-k'$ were coplanar, then $w \in \operatorname{int} Q$ would force $w = -a\hat{r} - bk'$ with $a, b > 0$, strictly between them. Hence the tangent cone is an embedded disk.
- *At $\gamma(a_e)$* (two fans and the adjacent cone sub-cones). The fans lie in $\{\langle \cdot, \gamma'\rangle \ge 0\}$, and the cone sub-cones in $\{\langle \cdot, \gamma'\rangle \le 0\}$ with $\langle p_v - \gamma(a_e), \gamma'\rangle < 0$. They share only the rays $\hat{r}_f \ne \hat{r}_g$ (E3′). The two cone sub-cones, $\operatorname{span}^+(\hat{r}_f, w)$ and $\operatorname{span}^+(w, \hat{r}_g)$ with $w = p_v - \gamma(a_e)$, do not fold onto each other, because $\hat{r}_f \ne \hat{r}_g$ and both are orthogonal to γ' while w is not.
- *At a junction crossing x of λ_v.* The tangent cone T_x of W_v at x is the union of two half-spaces, since W_v is a union of two convex sets near x. Let ℓ_1, ℓ_2 be the half-tangents of λ_v at x, one in each face of T_x.
  - The cone sub-cones near x consist of segments from p_v to points of λ_v. By strict star-shapedness these segments run into W_v, so the sub-cones lie in W_v near x. Their tangent sectors $\operatorname{span}^+(\ell_1, w)$ and $\operatorname{span}^+(w, \ell_2)$, with $w = p_v - x$, therefore lie in T_x. (This uses star-shapedness itself, not the convexity of T_x, which fails.)
  - The two sub-cones do not fold: projected along w, the half-tangents ℓ_1, ℓ_2 point in different directions, because the radial projection of λ_v is locally injective (each ray meets ∂W_v once, V1(a)) and λ_v is transverse to the rays.

– By the transversality in V1(b), the tangent sector of the outside piece lies in the closure of the complement of T_x.

So the sectors meet only along ℓ_1, w and ℓ_2, and the tangent cone is an embedded disk.

- *At the apex p_v.* h is homogeneous of degree one on each sector. The radial projection of λ_v to the unit sphere about p_v is injective (V1(a)). It is locally bi-Lipschitz: by transversality between corners, and by the no-fold arguments above at the corners. ϕ is bi-Lipschitz on each link segment. Hence h is bi-Lipschitz on the star of v.

Secant approximation. Take a triangulation of $|L|$ with these properties:

- it refines every cell of L and has every corner as a vertex;
- it is shape-regular off the sector cells;
- it is radial on the sector cells: its triangles there have the apex v as a vertex and a sub-segment of one link segment as the opposite side.

The two kinds of triangles meet only along those base segments. Let S be the secant map, linear on each triangle and equal to h at the vertices.

- *Off the sectors.* h is C^1 up to the boundary on each closed triangle. The standard per-triangle estimate gives $\mathrm{Lip}(S-h) \le C\,\omega(Dh, \mathrm{mesh})/\sin\theta_{\min}$, where ω is a modulus of continuity and $\theta_{\min}$ is bounded below by shape regularity.
- *On a radial triangle (v, y_0, y_1).* Its base lies on one link segment. Write $y = (1-u)y_0 + uy_1$ with $u \in [0,1]$, let $\ell = |y_1 - y_0|$, and let τ be the unit vector of the base; derivatives of e below are with respect to arc length ℓu.
  - A point $x = v + s(y-v)$ has barycentric coordinates $(1-s,\, s(1-u),\, su)$. So $S(x) = p_v + s(\hat{\phi}(y) - p_v)$, where $\hat{\phi}$ is the linear interpolation of ϕ on the base, and
  $$(S-h)(x) = s\,e(y), \qquad e = \hat{\phi} - \phi.$$
  This is exactly homogeneous of degree one in s.
  - Differentiating: $\partial_s x = y - v$ gives $\partial_s(S-h) = e(y)$. And $\partial_u x = s\ell\tau$ while $\partial_u(S-h) = s\ell e'$, so $D(S-h)\cdot\tau = e'$, with no division by the distance to the apex.
  - Hence $\|D(S-h)\| \le (|e|/|y-v| + |e'|)/\sin\beta$, where β is the angle between the ray $y-v$ and the base. β is bounded below by the fixed domain geometry, since the link is transverse to the rays.
  - Since $\hat{\phi}'$ is the mean of ϕ' over the base, $|e'| \le \omega(\phi', \mathrm{mesh})$ and $|e| \le \mathrm{mesh}\cdot\omega(\phi', \mathrm{mesh})$.
  - $D(S-h)$ is constant along rays, so the bound holds uniformly up to the apex, and $S-h$ is continuous at v. So $\mathrm{Lip}(S-h) \le C\,\omega(\phi', \mathrm{mesh}) \to 0$.

This uses only that ϕ is C^1 on each link segment; no curvature of λ_v is needed.

- *Conclusion.* With the intrinsic metric, $\mathrm{Lip}(S-h) \to 0$ and $|S-h| \to 0$ globally. For a fine enough triangulation:
  - $\mathrm{Lip}(S-h)$ is below the bi-Lipschitz constant of h, so S is injective on each shell, by $|S(x)-S(y)| \geq |h(x)-h(y)| - \mathrm{Lip}(S-h)\,|x-y|$;
  - h maps different shells to disjoint compact sets at some distance $\delta > 0$, so once $|S-h| < \delta/2$, S maps them to disjoint sets;
  - $|S-h| < m$, so $S(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$.

  S is PL, so it is a realization. □

Standard facts used without separate checking against a source: that the tangent cone of a C^1 regular arc is its tangent line (a ray at the end points); the implicit function theorem; gluing disks along boundary arcs; the per-triangle secant-approximation estimate for C^1 maps; bi-Lipschitz equivalence of the intrinsic and extrinsic metrics on a compact polyhedron; that a map whose tangent cones are embedded disks with positive opening angles is locally bi-Lipschitz; and that an injective, locally bi-Lipschitz map of a compact space is bi-Lipschitz.

Appendix C. Proof of Proposition 5.4

The idea is to take the re-trimmed surface Σ itself as the model of $|K|$, and the identity as a topological realization, then replace it by a fine secant approximation. Notation is that of Proposition 5.4.

- $X_f \colon U_f \to \mathbb{R}^3$ are the face parametrizations, and $D_f \subset U_f$ the compact parameter regions.
- $\phi \colon |K| \to \Sigma$ is the homeomorphism of C.

Appendix C.1. Margin

By C, ϕ maps each closed cell of K onto the corresponding closed cell of Σ. By B1–B3 each of these lies in the zone of its cell. Closed cells are compact and finitely many, and zones are open. So there is $m > 0$ such that the m-neighbourhood of every closed cell of Σ lies in its zone.

Appendix C.2. A curved triangulation of Σ

Fix a small $h > 0$.

1. *Edge vertices.* On each curve $\tilde{\gamma}_e$, fix one C^2 regular parametrization (by arc length, say), and choose vertices on it at 3D spacing about h, including its end points. A closed edge gets at least three vertices. These choices are made once, in $\mathbb{R}^3$, and are shared by all faces containing e (both sides of a seam).
2. *Cutting along the edge graph.* In each D_f, pull back the edge vertices by X_f; this is possible because X_f is injective on D_f. Cut D_f along its edge graph G_f (R), seams included. Note that $\partial D_f \subseteq G_f$: a point of ∂D_f off G_f would be mapped by X_f into the open face, which by C is an open surface homeomorphic, through X_f^{-1}, to an open

subset of U_f; by invariance of domain such a point is interior to D_f. The result of cutting is a finite union of compact regions, each bounded by arcs of G_f (a seam arc bounds a region on each of its two sides). A region need not be simply connected: faces with inner wires give multiply connected regions. Join consecutive edge vertices along each arc by chords. For small h each boundary component of a region becomes a simple closed polygon, these polygons are disjoint, and they bound a polygonal region close to the original one, by R: the arcs are simple C^2 regular, and the sector angles lie in $(0, 2\pi)$.

3. *Triangulation.* Triangulate each chord region shape-regularly, with no extra vertices on its boundary. At every vertex of G_f, including smooth passages such as seam vertices, put one interior vertex near it in each adjacent sector and join it to the vertex, so that no triangle has two curved edges. The shape-regularity constant depends on the smallest sector angle.
4. *Curved triangles.* Replace each chord by its arc. An arc deviates from its chord by $o(h) \cdot h$, which is much smaller than $h \sin \theta_{\min}$, so the curved triangles tile D_f, and every arc of G_f, seams included, is a union of edges of the triangulation.

Let T be the resulting abstract complex. For small h it is a simplicial complex, and it subdivides the cells of Σ, hence, through ϕ, the cells of K. We metrize $|T|$ by standard (equilateral) simplices and use the intrinsic length metric on each shell.

The map $h_0 \colon |T| \to \Sigma$.

- On a triangle with one curved edge, use Lemma Appendix A.1, with the curved edge parametrized by the pull-back of the fixed parametrization of $\tilde{\gamma}_e$. Precompose with the affine map from the standard simplex, and compose with X_f.
- On a straight triangle, use the affine map, composed with X_f.

Then:

- h_0 *agrees on shared edges.*
  - An interior straight edge is affine in U_f on both sides.
  - An edge on $\tilde{\gamma}_e$ between faces f and g is $s \mapsto \tilde{\gamma}_e(t_0 + s(t_1 - t_0))$ from both sides, by Lemma Appendix A.1(a).
  - A seam edge lies inside D_f, and both sides use the same pull-back of the fixed parametrization of the seam curve, so it is mapped the same way from both sides.
- h_0 *is a homeomorphism.* It is a homeomorphism onto Σ, by Lemma Appendix A.1(d) and the injectivity of X_f on D_f.
- *Regularity.* On each closed triangle h_0 is C^1 up to the boundary, with injective differential. This follows from Lemma Appendix A.1(c),(d), since X_f is C^2 regular on a neighbourhood of D_f.

Appendix C.3. h_0 is bi-Lipschitz on each shell

On a shell, $|T|$ is compact and h_0 is injective, so it suffices that h_0 is locally bi-Lipschitz.

- *Inside a face,* including across interior edges and at interior vertices of T: the map to D_f is a piecewise-C^1 homeomorphism with invertible differential on each closed piece. Its tangent map at a vertex is a sectorwise linear, orientation-preserving map of the plane of degree one, hence a homeomorphism. Composing with the regular parametrization X_f keeps it locally bi-Lipschitz.

- *Along an edge (seams included), away from its end points:* by T the two inward tangent half-planes are distinct. For a seam they are opposite in the tangent plane of S_f. The angle between them is continuous and positive, hence bounded away from 0 on compact subarcs; near the end points the vertex argument below takes over. So the tangent cone at each point is a union of two half-planes that meet only along their common line. This also covers vertices of T that lie on $\tilde{\gamma}_e$: there the triangle sectors on each side tile the two half-planes.

- *At a vertex c_v:* by T the tangent cone of Σ, the union of the tangent sectors of the edge graphs at the preimages of c_v, is an embedded disk; by R each sector has an opening angle in $(0, 2\pi)$. h_0 is C^1 up to c_v on every closed triangle, so the blow-up argument applies: rescale at c_v, pass to the tangent cone, and use C^1 convergence on each sector.

Appendix C.4. Secant approximation

Refine T by affine subdivision of each simplex; this keeps shape regularity and keeps h_0 C^1 on each original closed triangle. Let S be the secant map: linear on each refined triangle and equal to h_0 at the vertices.

- *The estimate.* On each triangle, $\mathrm{Lip}(S - h_0) \le C\,\omega(Dh_0, \mathrm{mesh})/\sin\theta_{\min}$ and $|S - h_0| \le \mathrm{mesh} \cdot \omega(Dh_0, \mathrm{mesh})$, where ω is a modulus of continuity. In the intrinsic metric of each shell these bounds are global.

- *Within a shell.* Once $\mathrm{Lip}(S - h_0)$ is below the bi-Lipschitz constant of h_0, S is injective there.

- *Between shells.* h_0 maps different shells to disjoint compact sets at distance $\delta > 0$. Once $|S - h_0| < \delta/2$, S maps them to disjoint sets.

- *Zones.* Once $|S - h_0| < m$: every point of a closed cell lies on a simplex of T inside that cell, so $S(\bar{\sigma}) \subseteq \mathcal{Z}(\sigma)$ for vertices, edges and faces. In particular $S(\mathrm{int}\,\sigma) \subseteq \mathcal{Z}(\sigma)$.

$S \circ h_0^{-1} \circ \phi$ is then a PL embedding of $|K|$, linear on the simplices of a triangulation that subdivides the cells of K, with every cell in its zone. That is a realization. □

Standard facts used without separate checking against a source:

- existence of shape-regular triangulations of a polygonal region, possibly multiply connected, with prescribed boundary vertices and corner fans as described;

- the per-triangle secant-approximation estimate;

- that a sectorwise linear, orientation-preserving map of the plane of degree one is a homeomorphism;
- that a map that is C^1 up to the vertex on each sector, with an embedded tangent cone of positive opening angles, is locally bi-Lipschitz (blow-up);
- that an injective, locally bi-Lipschitz map of a compact space is bi-Lipschitz.

Appendix D. Formal verification in Lean

The following are proved in Lean 4 with Mathlib (repository directory `lean/`). The build has no `sorry`, and the results depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Lean statement	Content
`inner_eq_zero_of_conical`	at a nearest point x, $\langle p-x, x-a\rangle = 0$ (Lemma 4.2)
`inner_sub_eq_norm_sq_of_conical`	$\langle p-x, p-a\rangle = \lvert p-x\rvert^2$
`not_isCritical_of_conical`	p is not critical (Lemma 4.2)
`not_isCritical_of_conicalCover`	pointwise form of Corollary 4.3
`carrier_mono`, `carrier_eq_zone_of_mono`	carriers are monotone, and equal the zones when the zones are augmented
`carried_of_realization`, `image_closedCell_subset_carrier`	realizations are carried
`carried_straightLine`, `continuous_straightLine`	Lemma 3.1 when the carriers are convex

The statements are for an arbitrary real inner product space. Criticality is formalized as $0 \in \mathrm{conv}\{p-x : x \text{ nearest}\}$.

How the Lean statements correspond to the paper.

- The hypothesis `LocallyConicalAt M a r` is radial closure of M inside the open ball $B(a,r)$. The paper's definition of $\rho(a)$ implies it for $r < \rho(a)$; the two are not the same statement.
- Lean's `Carried` is pointwise on open cells. By `carrier_mono` it agrees with the paper's closed-cell definition.
- Positivity of $\lambda(M)$ for polyhedra (Corollary 4.3) is proved on paper only.

References

[1] L.-E. Andersson, T. J. Peters, N. F. Stewart, Equivalence of topological form for curvilinear geometric objects, *Int. J. Comput. Geom. Appl.* 10(6) (2000) 609–622.

[2] L.-E. Andersson, N. F. Stewart, M. Zidani, Error analysis for operations in solid modeling in the presence of uncertainty, *SIAM J. Sci. Comput.* 29(2) (2007) 811–826.

[3] L.-E. Andersson, N. F. Stewart, M. Zidani, Conditions for use of a non-selfintersection conjecture, *Comput. Aided Geom. Design* 23(7) (2006) 599–611.

[4] F. Chazal, D. Cohen-Steiner, A condition for isotopic approximation, *Graphical Models* 67(5) (2005) 390–404.

[5] F. Chazal, D. Cohen-Steiner, A. Lieutier, A sampling theory for compact sets in Euclidean space, *Discrete Comput. Geom.* 41(3) (2009) 461–479.

[6] F. Chazal, D. Cohen-Steiner, A. Lieutier, Normal cone approximation and offset shape isotopy, *Comput. Geom.* 42(6–7) (2009) 566–581.

[7] F. Chazal, A. Lieutier, Stability and computation of topological invariants of solids in $\mathbb{R}^n$, *Discrete Comput. Geom.* 37 (2007) 601–617.

[8] S. di Rocco, D. Eklund, O. Gäfvert, Computing geometric feature sizes for algebraic manifolds, arXiv:2209.01654.

[9] A. H. Durfee, Neighborhoods of algebraic sets, *Trans. Amer. Math. Soc.* 276(2) (1983) 517–530.

[10] H. Federer, Curvature measures, *Trans. Amer. Math. Soc.* 93(3) (1959) 418–491.

[11] J. H. G. Fu, Tubular neighborhoods in Euclidean spaces, *Duke Math. J.* 52(4) (1985) 1025–1046.

[12] D. J. Jackson, Boundary representation modelling with local tolerances, in *Proc. 3rd ACM Symp. Solid Modeling and Applications (SMA '95)*, 1995, 247–254.

[13] V. J. Milenkovic, Verifiable implementations of geometric algorithms using finite precision arithmetic, *Artificial Intelligence* 37(1–3) (1988) 377–401.

[14] E. E. Moise, *Geometric Topology in Dimensions 2 and 3*, Graduate Texts in Mathematics 47, Springer, New York, 1977.

[15] Open Cascade, *OpenCascade Technology documentation*, version 8.0.1: BRep format specification, §§7.2–7.4 (shape tolerances).

[16] J. Qi, V. Shapiro, ε-topological formulation of tolerant solid modeling, *Computer-Aided Design* 38(4) (2006) 367–377.

[17] J. Qi, V. Shapiro, Geometric interoperability with epsilon solidity, *J. Comput. Inf. Sci. Eng.* 6(3) (2006) 213–220.

[18] T. Sakkalis, G. Shen, N. M. Patrikalakis, Topological and geometric properties of interval solid models, *Graphical Models* 63(3) (2001) 163–175.

[19] T. Sakkalis, T. J. Peters, J. Bisceglio, Isotopic approximations and interval solids, *Computer-Aided Design* 36 (2004) 1089–1100.

[20] M. Segal, Using tolerances to guarantee valid polyhedral modeling results, *Computer Graphics (Proc. SIGGRAPH '90)* 24(4) (1990) 105–114.

[21] V. Shapiro, Homotopy conditions for tolerant geometric queries, in *Reliable Implementation of Real Number Algorithms*, LNCS 5045, Springer, 2008, 162–180.

[22] N. F. Stewart, M. Zidani, Transfinite interpolation for well-definition in error analysis in solid modelling, in *Reliable Implementation of Real Number Algorithms*, LNCS 5045, Springer, 2008, 181–192.

[23] F. Waldhausen, On irreducible 3-manifolds which are sufficiently large, *Ann. of Math.* 87 (1968) 56–88.

[24] Code, proofs and data repository; link withheld for double-anonymized review (public repository).